

Recursive Formulae for Trotter-Suzuki Decompositions

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Introduction

In this document we gather the results of our derivations within the Mathematica notebook (Efficient_Trotterizations.nb). First we define our basis for each order n , which is written in terms of right-nested commutators as described by Arnal *et al.* (2021). We also symmetrize the basis according to the symmetry $A \leftrightarrow B$, which allows us to derive the recursive formulae for one of the steps only (we choose the step with the A operator). For orders $n \geq 6$ there also exist identities, which help reduce the total number of basis commutators. We also write down these relations.

In the second part we present the recursive formulae for the scheme coefficients $\alpha_A^{(i,i-1)}$, $\beta_A^{(i,i-1)}$, $\gamma_{j,A}^{(i,i-1)}$, $\dots$. These are written down in terms of the step done with the commutator A . By replacing the parameters a_i with b_i and the scheme coefficients with their respective counterparts, it is possible to find the formulae for the step in the B operator. In this formulation the coefficients ν and σ simply add up the scheme parameters as such:

$$\nu^{(i)} = \nu^{(i-1)} + a_i, \tag{1}$$

$$\sigma^{(i-1)} = \sigma^{(i-2)} + b_i. \tag{2}$$

Basis

2nd order

$$C_1 = [A^2B], \quad \tilde{\alpha} = -\frac{1}{24},$$

$$C_2 = [B^2A], \quad \tilde{\beta} = \frac{1}{12}.$$

4th order

$$D_1 = [A^4B] = [A^2C_1], \quad \tilde{\gamma}_1 = \frac{7}{5760},$$

$$D_2 = [A^2BAB] = -[A^2C_2] = [ABA^2B] = [ABC_1], \quad \tilde{\gamma}_2 = \frac{1}{480},$$

$$D_3 = [BA^3B] = [BAC_1], \quad \tilde{\gamma}_3 = \frac{1}{360},$$

$$D_4 = [AB^3A] = [ABC_2], \quad \tilde{\gamma}_4 = \frac{1}{360},$$

$$D_5 = [B^2ABA] = -[B^2C_1] = [BAB^2A] = [BAC_2], \quad \tilde{\gamma}_5 = -\frac{1}{120},$$

$$D_6 = [B^4A] = [B^2C_2], \quad \tilde{\gamma}_6 = -\frac{1}{720}.$$

6th order

$$E_1 = [A^6B] = [A^2D_1], \quad \tilde{\delta}_1 = -\frac{31}{967680},$$

$$E_2 = [A^2BA^3B] = [A^2D_3], \quad \tilde{\delta}_2 = -\frac{19}{80640},$$

$$E_3 = [ABA^4B] = [ABD_1], \quad \tilde{\delta}_3 = \frac{23}{161280},$$

$$E_4 = [BA^5B] = [BAD_1], \quad \tilde{\delta}_4 = -\frac{1}{10080},$$

$$E_5 = [A^2BABAB] = [A^2B^2A^2B] = -[A^2D_5], \quad \tilde{\delta}_5 = -\frac{1}{10080},$$

$$E_6 = [ABA^2BAB] = [ABABA^2B] = [ABD_2], \quad \tilde{\delta}_6 = -\frac{1}{20160},$$

$$E_7 = [AB^2A^3B] = [ABD_3], \quad \tilde{\delta}_7 = \frac{13}{60480},$$

$$E_8 = [BABA^3B] = [BAD_3], \quad \tilde{\delta}_8 = -\frac{1}{5040},$$

$$E_9 = [BA^3BAB] = [BA^2BA^2B] = [BAD_2], \quad \tilde{\delta}_9 = -\frac{1}{3360},$$

$$\begin{aligned}
E_{10} &= [AB^3ABA] = [AB^2AB^2A] = [ABD_5], & \tilde{\delta}_{10} &= -\frac{19}{40320}, \\
E_{11} &= [ABAB^3A] = [ABD_4], & \tilde{\delta}_{11} &= \frac{1}{6720}, \\
E_{12} &= [BA^2B^3A] = [BAD_4], & \tilde{\delta}_{12} &= -\frac{1}{7560}, \\
E_{13} &= [BAB^2ABA] = [BABAB^2A] = [BAD_5], & \tilde{\delta}_{13} &= \frac{1}{1008}, \\
E_{14} &= [B^2ABABA] = [B^2A^2B^2A] = -[B^2D_2], & \tilde{\delta}_{14} &= -\frac{1}{10080}, \\
E_{15} &= [AB^5A] = [ABD_6], & \tilde{\delta}_{15} &= -\frac{1}{10080}, \\
E_{16} &= [BAB^4A] = [BAD_6], & \tilde{\delta}_{16} &= \frac{1}{10080}, \\
E_{17} &= [B^2AB^3A] = [B^2D_4], & \tilde{\delta}_{17} &= \frac{1}{5040}, \\
E_{18} &= [B^6A] = [B^2D_6], & \tilde{\delta}_{18} &= \frac{1}{30240}.
\end{aligned}$$

$$\begin{aligned}
[A^4BAB] &= [A^3BA^2B] = [A^2D_2] = 2E_2 - E_3, \\
[A^3B^2AB] &= -[A^2D_4] = 3E_5 - 3E_6 + E_7, \\
[B^2A^4B] &= [B^2D_1] = 2E_8 - E_9, \\
[A^2B^4A] &= [A^2D_6] = 2E_{11} - E_{10}, \\
[B^3A^2BA] &= -[B^2D_3] = 3E_{14} - 3E_{13} + E_{12}, \\
[B^4ABA] &= [B^3AB^2A] = [B^2D_5] = 2E_{17} - E_{16}.
\end{aligned}$$

8th order

$$\begin{aligned}
F_1 &= [A^8B] = [A^2E_1], & \tilde{\epsilon}_1 &= \frac{127}{154828800}, \\
F_2 &= [A^3BA^4B] = [A^2E_3], & \tilde{\epsilon}_2 &= \frac{11}{774144}, \\
F_3 &= [A^2BA^5B] = [A^2E_4], & \tilde{\epsilon}_3 &= -\frac{41}{3225600}, \\
F_4 &= [ABA^6B] = [ABE_1], & \tilde{\epsilon}_4 &= \frac{17}{9676800}, \\
F_5 &= [BA^7B] = [BAE_1], & \tilde{\epsilon}_5 &= \frac{1}{302400}, \\
F_6 &= [A^3B^2A^3B] = [A^2E_7], & \tilde{\epsilon}_6 &= \frac{1}{290304}, \\
F_7 &= [A^2BABA^3B] = [A^2E_8], & \tilde{\epsilon}_7 &= -\frac{1}{48384},
\end{aligned}$$

$$\begin{aligned}
F_8 &= [A^2BA^3BAB] = [A^2BA^2BA^2B] = [A^2E_9], & \tilde{\epsilon}_8 &= \frac{1}{26880}, \\
F_9 &= [ABA^2BA^3B] = [ABE_2], & \tilde{\epsilon}_9 &= -\frac{1}{13824}, \\
F_{10} &= [ABABA^4B] = [ABE_3], & \tilde{\epsilon}_{10} &= \frac{1}{15360}, \\
F_{11} &= [AB^2A^5B] = [ABE_4], & \tilde{\epsilon}_{11} &= -\frac{23}{1612800}, \\
F_{12} &= [BA^3BA^3B] = [BAE_2], & \tilde{\epsilon}_{12} &= \frac{211}{4838400}, \\
F_{13} &= [BA^2BA^4B] = [BAE_3], & \tilde{\epsilon}_{13} &= -\frac{37}{1209600}, \\
F_{14} &= [BABA^5B] = [BAE_4], & \tilde{\epsilon}_{14} &= \frac{7}{691200}, \\
F_{15} &= [A^2BA^2B^2AB] = -[A^2E_{12}], & \tilde{\epsilon}_{15} &= \frac{47}{1209600}, \\
F_{16} &= [A^2BAB^2A^2B] = [A^2BABABAB] = -[A^2E_{13}], & \tilde{\epsilon}_{16} &= -\frac{281}{2419200}, \\
F_{17} &= [A^2B^2ABA^2B] = [A^2B^2A^2BAB] = -[A^2E_{14}], & \tilde{\epsilon}_{17} &= \frac{163}{1612800}, \\
F_{18} &= [ABA^2BABAB] = [ABA^2B^2A^2B] = [ABE_5], & \tilde{\epsilon}_{18} &= \frac{73}{1612800}, \\
F_{19} &= [ABABA^2BAB] = [ABABABA^2B] = [ABE_6], & \tilde{\epsilon}_{19} &= -\frac{43}{2419200}, \\
F_{20} &= [ABAB^2A^3B] = [ABE_7], & \tilde{\epsilon}_{20} &= -\frac{79}{1209600}, \\
F_{21} &= [AB^2ABA^3B] = [ABE_8], & \tilde{\epsilon}_{21} &= -\frac{1}{193536}, \\
F_{22} &= [AB^2A^3BAB] = [AB^2A^2BA^2B] = [ABE_9], & \tilde{\epsilon}_{22} &= -\frac{13}{3225600}, \\
F_{23} &= [BA^3BABAB] = [BA^3B^2A^2B] = [BAE_5], & \tilde{\epsilon}_{23} &= -\frac{1}{25200}, \\
F_{24} &= [BA^2BA^2BAB] = [BA^2BABA^2B] = [BAE_6], & \tilde{\epsilon}_{24} &= \frac{1}{11200}, \\
F_{25} &= [BA^2B^2A^3B] = [BAE_7], & \tilde{\epsilon}_{25} &= -\frac{1}{75600}, \\
F_{26} &= [BABABA^3B] = [BAE_8], & \tilde{\epsilon}_{26} &= \frac{1}{12600}, \\
F_{27} &= [BABA^3BAB] = [BABA^2BA^2B] = [BAE_9], & \tilde{\epsilon}_{27} &= -\frac{1}{25200}, \\
F_{28} &= [B^2ABA^4B] = [B^2E_3], & \tilde{\epsilon}_{28} &= -\frac{1}{67200},
\end{aligned}$$

$$\begin{aligned}
F_{29} &= [A^2BAB^4A] = [A^2E_{16}], & \tilde{\epsilon}_{29} &= -\frac{1}{67200}, \\
F_{30} &= [ABAB^3ABA] = [ABAB^2AB^2A] = [ABE_{10}], & \tilde{\epsilon}_{30} &= -\frac{31}{403200}, \\
F_{31} &= [ABABAB^3A] = [ABE_{11}], & \tilde{\epsilon}_{31} &= \frac{11}{201600}, \\
F_{32} &= [AB^2A^2B^3A] = [ABE_{12}], & \tilde{\epsilon}_{32} &= -\frac{1}{1209600}, \\
F_{33} &= [AB^2AB^2ABA] = [AB^2ABAB^2A] = [ABE_{13}], & \tilde{\epsilon}_{33} &= \frac{17}{134400}, \\
F_{34} &= [AB^3ABABA] = [AB^3A^2B^2A] = [ABE_{14}], & \tilde{\epsilon}_{34} &= -\frac{1}{19200}, \\
F_{35} &= [BA^2B^3ABA] = [BA^2B^2AB^2A] = [BAE_{10}], & \tilde{\epsilon}_{35} &= \frac{31}{604800}, \\
F_{36} &= [BA^2BAB^3A] = [BAE_{11}], & \tilde{\epsilon}_{36} &= -\frac{1}{60480}, \\
F_{37} &= [BABA^2B^3A] = [BAE_{12}], & \tilde{\epsilon}_{37} &= -\frac{1}{50400}, \\
F_{38} &= [BABAB^2ABA] = [BABABAB^2A] = [BAE_{13}], & \tilde{\epsilon}_{38} &= -\frac{1}{5400}, \\
F_{39} &= [BAB^2ABABA] = [BAB^2A^2B^2A] = [BAE_{14}], & \tilde{\epsilon}_{39} &= \frac{1}{12600}, \\
F_{40} &= [B^2A^2BAB^2A] = [B^2A^2B^2ABA] = -[B^2E_5], & \tilde{\epsilon}_{40} &= \frac{1}{12600}, \\
F_{41} &= [B^2ABA^2B^2A] = [B^2ABABABA] = -[B^2E_6], & \tilde{\epsilon}_{41} &= -\frac{11}{151200}, \\
F_{42} &= [B^2AB^2A^2BA] = -[B^2E_7], & \tilde{\epsilon}_{42} &= \frac{1}{100800}, \\
F_{43} &= [ABAB^5A] = [ABE_{15}], & \tilde{\epsilon}_{43} &= -\frac{29}{1209600}, \\
F_{44} &= [AB^2AB^4A] = [ABE_{16}], & \tilde{\epsilon}_{44} &= \frac{83}{1209600}, \\
F_{45} &= [AB^3AB^3A] = [ABE_{17}], & \tilde{\epsilon}_{45} &= -\frac{31}{1209600}, \\
F_{46} &= [BA^2B^5A] = [BAE_{15}], & \tilde{\epsilon}_{46} &= \frac{1}{50400}, \\
F_{47} &= [BABAB^4A] = [BAE_{16}], & \tilde{\epsilon}_{47} &= -\frac{1}{10080}, \\
F_{48} &= [BAB^2AB^3A] = [BAE_{17}], & \tilde{\epsilon}_{48} &= \frac{1}{30240}, \\
F_{49} &= [B^2AB^3ABA] = [B^2AB^2AB^2A] = [B^2E_{10}], & \tilde{\epsilon}_{49} &= -\frac{1}{20160},
\end{aligned}$$

$$\begin{aligned}
F_{50} &= [B^2ABAB^3A] = [B^2E_{11}], & \tilde{\epsilon}_{50} &= \frac{1}{15120}, \\
F_{51} &= [B^3A^2B^3A] = [B^2E_{12}], & \tilde{\epsilon}_{51} &= -\frac{1}{90720}, \\
F_{52} &= [AB^7A] = [ABE_{18}], & \tilde{\epsilon}_{52} &= \frac{1}{302400}, \\
F_{53} &= [BAB^6A] = [BAE_{18}], & \tilde{\epsilon}_{53} &= -\frac{1}{75600}, \\
F_{54} &= [B^2AB^5A] = [B^2E_{15}], & \tilde{\epsilon}_{54} &= \frac{1}{50400}, \\
F_{55} &= [B^3AB^4A] = [B^2E_{16}], & \tilde{\epsilon}_{55} &= -\frac{1}{60480}, \\
F_{56} &= [B^8A] = [B^2E_{18}], & \tilde{\epsilon}_{56} &= -\frac{1}{1209600}.
\end{aligned}$$

$$\begin{aligned}
[A^4BA^3B] &= [A^2E_2] = 3F_2 - 3F_3 + F_4, \\
[A^6BAB] &= [A^5BA^2B] = [A^4D_2] = 5F_2 - 6F_3 + 2F_4, \\
[A^4BABAB] &= [A^4B^2A^2B] = [A^2E_5] = 4F_6 - 14F_7 + 9F_8 - 16F_9 + 22F_{10} \\
&\quad - 4F_{11} + 9F_{12} - 12F_{13} + 3F_{14}, \\
[A^3BA^2BAB] &= [A^3BABA^2B] = [A^2E_6] = F_6 - 4F_7 + 4F_8 - 6F_9 + 7F_{10} \\
&\quad - F_{11} + 3F_{12} - 4F_{13} + F_{14}, \\
[B^2A^6B] &= [B^2E_1] = F_{12} - 3F_{13} + 3F_{14}, \\
[A^5B^2AB] &= -[A^4D_4] = 10F_6 - 30F_7 + 15F_8 - 30F_9 + 45F_{10} \\
&\quad - 9F_{11} + 18F_{12} - 24F_{13} + 6F_{14}, \\
[A^2B^2A^4B] &= [A^2B^2D_1] = 2F_7 - F_8, \\
[ABA^3BA^2B] &= [ABA^4BAB] = [ABA^2D_2] = 2F_9 - F_{10}, \\
[BA^5BAB] &= [BA^4BA^2B] = [BA^3D_2] = 2F_{12} - F_{13}, \\
[A^3B^3A^2B] &= [A^3B^2ABAB] = -[A^2E_{10}] = 2F_{16} - 2F_{19} + F_{22}, \\
[A^3BAB^2AB] &= -[A^2E_{11}] = 4F_{15} - 5F_{16} + 3F_{17} - 3F_{18} \\
&\quad + 5F_{19} - 4F_{20} + F_{21}, \\
[B^2A^2BA^3B] &= [B^2E_2] = -F_{15} + 4F_{16} - 3F_{17} - 3F_{18} + 2F_{19} \\
&\quad + 2F_{20} - F_{22} + 2F_{23} - 3F_{24} \\
&\quad - 2F_{26} + 3F_{27} + F_{28}, \\
[B^3A^5B] &= [B^2E_4] = -2F_{15} + 8F_{16} - 6F_{17} - 6F_{18} + 4F_{19} \\
&\quad + 4F_{20} - 2F_{22} + 3F_{23} - 3F_{24} \\
&\quad + F_{25} - 6F_{26} + 3F_{27} + 3F_{28},
\end{aligned}$$

$$\begin{aligned}
[A^4B^3AB] &= -[A^4D_6] = 8F_{15} - 12F_{16} + 6F_{17} - 6F_{18} \\
&\quad + 12F_{19} - 8F_{20} + 2F_{21} - F_{22}, \\
[A^2B^3A^3B] &= [A^2B^2D_3] = F_{15} - 3F_{16} + 3F_{17}, \\
[ABA^3B^2AB] &= -[ABA^2D_4] = 3F_{18} - 3F_{19} + F_{20}, \\
[AB^3A^4B] &= [AB^3D_1] = 2F_{21} - F_{22}, \\
[BA^4B^2AB] &= -[BA^3D_4] = 3F_{23} - 3F_{24} + F_{25}, \\
[BAB^2A^4B] &= [BAB^2D_1] = 2F_{26} - F_{27}, \\
[B^2A^4BAB] &= [B^2A^3BA^2B] = [B^2A^2D_2] = -2F_{15} + 8F_{16} - 6F_{17} - 6F_{18} + 4F_{19} \\
&\quad + 4F_{20} - 2F_{22} + 4F_{23} - 6F_{24} \\
&\quad - 4F_{26} + 6F_{27} + F_{28}, \\
[A^2B^4ABA] &= [A^2B^3AB^2A] = [A^2B^2D_5] = -2F_{42} + 8F_{41} - 6F_{40} - 6F_{39} + 4F_{38} \\
&\quad + 4F_{37} - 2F_{35} + 4F_{34} - 6F_{33} \\
&\quad - 4F_{31} + 6F_{30} + F_{29}, \\
[ABA^2B^4A] &= [ABA^2D_6] = 2F_{31} - F_{30}, \\
[AB^4A^2BA] &= -[AB^3D_3] = 3F_{34} - 3F_{33} + F_{32}, \\
[BA^3B^4A] &= [BA^3D_6] = 2F_{36} - F_{35}, \\
[BAB^3A^2BA] &= -[BAB^2D_3] = 3F_{39} - 3F_{38} + F_{37}, \\
[B^2A^3B^3A] &= [B^2A^2D_4] = F_{42} - 3F_{41} + 3F_{40}, \\
[B^4A^3BA] &= -[B^4D_1] = 8F_{42} - 12F_{41} + 6F_{40} - 6F_{39} \\
&\quad + 12F_{38} - 8F_{37} + 2F_{36} - F_{35}, \\
[A^3B^5A] &= [A^2E_{15}] = -2F_{42} + 8F_{41} - 6F_{40} - 6F_{39} + 4F_{38} \\
&\quad + 4F_{37} - 2F_{35} + 3F_{34} - 3F_{33} \\
&\quad + F_{32} - 6F_{31} + 3F_{30} + 3F_{29}, \\
[A^2B^2AB^3A] &= [A^2E_{17}] = -F_{42} + 4F_{41} - 3F_{40} - 3F_{39} + 2F_{38} \\
&\quad + 2F_{37} - F_{35} + 2F_{34} - 3F_{33} \\
&\quad - 2F_{31} + 3F_{30} + F_{29}, \\
[B^3ABA^2BA] &= -[B^2E_8] = 4F_{42} - 5F_{41} + 3F_{40} - 3F_{39} \\
&\quad + 5F_{38} - 4F_{37} + F_{36}, \\
[B^3A^3B^2A] &= [B^3A^2BABA] = -[B^2E_9] = 2F_{41} - 2F_{38} + F_{35}, \\
[AB^5ABA] &= [AB^4AB^2A] = [AB^3D_5] = 2F_{45} - F_{44}, \\
[BAB^3AB^2A] &= [BAB^4ABA] = [BAB^2D_5] = 2F_{48} - F_{47}, \\
[B^2A^2B^4A] &= [B^2A^2D_6] = 2F_{50} - F_{49},
\end{aligned}$$

$$\begin{aligned}
[B^5 A^2 B A] &= -[B^4 D_3] = 10F_{51} - 30F_{50} + 15F_{49} - 30F_{48} + 45F_{47} \\
&\quad - 9F_{46} + 18F_{45} - 24F_{44} + 6F_{43}, \\
[A^2 B^6 A] &= [A^2 E_{18}] = F_{45} - 3F_{44} + 3F_{43}, \\
[B^3 A B^2 A B A] &= [B^3 A B A B^2 A] = [B^2 E_{13}] = F_{51} - 4F_{50} + 4F_{49} - 6F_{48} + 7F_{47} \\
&\quad - F_{46} + 3F_{45} - 4F_{44} + F_{43}, \\
[B^4 A B A B A] &= [B^4 A^2 B^2 A] = [B^2 E_{14}] = 4F_{51} - 14F_{50} + 9F_{49} - 16F_{48} + 22F_{47} \\
&\quad - 4F_{46} + 9F_{45} - 12F_{44} + 3F_{43}, \\
[B^6 A B A] &= [B^5 A B^2 A] = [B^4 D_5] = 5F_{55} - 6F_{54} + 2F_{53}, \\
[B^4 A B^3 A] &= [B^2 E_{17}] = 3F_{55} - 3F_{54} + F_{53}.
\end{aligned}$$

Coefficient recursive formulae

2nd order

$$\alpha_A^{(i,i-1)} = \alpha_B^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \sigma^{(i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \sigma^{(i-1)}, \quad (3)$$

$$\beta_A^{(i,i-1)} = \beta_B^{(i-1,i-1)} + \tilde{\beta} a_i (\sigma^{(i-1)})^2. \quad (4)$$

4th order

$$\begin{aligned}
\gamma_{1,A}^{(i,i-1)} &= \gamma_{1,B}^{(i-1,i-1)} + \tilde{\gamma}_1 a_i^4 \sigma^{(i-1)} + (\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \sigma^{(i-1)} - (\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 (\nu^{(i-1)})^2 \sigma^{(i-1)} \\
&\quad - \tilde{\gamma}_6 a_i (\nu^{(i-1)})^3 \sigma^{(i-1)} + \tilde{\alpha} a_i^2 \alpha_B^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \alpha_B^{(i-1,i-1)},
\end{aligned} \quad (5)$$

$$\begin{aligned}
\gamma_{2,A}^{(i,i-1)} &= \gamma_{2,B}^{(i-1,i-1)} + \tilde{\gamma}_2 a_i^3 (\sigma^{(i-1)})^2 - (2\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} (\sigma^{(i-1)})^2 - 2\tilde{\gamma}_6 a_i (\nu^{(i-1)})^2 (\sigma^{(i-1)})^2 \\
&\quad - \tilde{\alpha} a_i^2 \beta_B^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \quad (6)$$

$$\begin{aligned}
\gamma_{3,A}^{(i,i-1)} &= \gamma_{3,B}^{(i-1,i-1)} + \tilde{\gamma}_3 a_i^3 (\sigma^{(i-1)})^2 - \tilde{\gamma}_5 a_i^2 \nu^{(i-1)} (\sigma^{(i-1)})^2 - \tilde{\gamma}_6 a_i (\nu^{(i-1)})^2 (\sigma^{(i-1)})^2 \\
&\quad - 2\tilde{\beta} a_i \sigma^{(i-1)} \alpha_B^{(i-1,i-1)},
\end{aligned} \quad (7)$$

$$\gamma_{4,A}^{(i,i-1)} = \gamma_{4,B}^{(i-1,i-1)} + \tilde{\gamma}_4 a_i^2 (\sigma^{(i-1)})^3 + \tilde{\gamma}_6 a_i \nu^{(i-1)} (\sigma^{(i-1)})^3 + \tilde{\beta} a_i \sigma^{(i-1)} \beta_B^{(i-1,i-1)}, \quad (8)$$

$$\gamma_{5,A}^{(i,i-1)} = \gamma_{5,B}^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 (\sigma^{(i-1)})^3 + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} (\sigma^{(i-1)})^3 - 2\tilde{\beta} a_i \sigma^{(i-1)} \beta_B^{(i-1,i-1)}, \quad (9)$$

$$\gamma_{6,A}^{(i,i-1)} = \gamma_{6,B}^{(i-1,i-1)} + \tilde{\gamma}_6 a_i (\sigma^{(i-1)})^4. \quad (10)$$

6th order

$$\begin{aligned}
\delta_{1,A}^{(i,i-1)} &= \delta_{1,B}^{(i-1,i-1)} + \tilde{\delta}_1 a_i^6 \sigma^{(i-1)} + \left(\tilde{\delta}_2 + \tilde{\delta}_3 + \tilde{\delta}_4 \right) a_i^5 \nu^{(i-1)} \sigma^{(i-1)} \\
&+ \left(\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \\
&- \left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} - \left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \\
&- \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^5 \sigma^{(i-1)} + \tilde{\gamma}_1 a_i^4 \alpha_B^{(i-1,i-1)} + (\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&- (\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \gamma_{1,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned} \tag{11}$$

$$\begin{aligned}
\delta_{2,A}^{(i,i-1)} &= \delta_{2,B}^{(i-1,i-1)} + \tilde{\delta}_2 a_i^5 \left(\sigma^{(i-1)} \right)^2 + \left(3\tilde{\delta}_5 + 2\tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + 2\tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \\
&- \left(3\tilde{\delta}_{10} + 4\tilde{\delta}_{11} + 4\tilde{\delta}_{12} + 3\tilde{\delta}_{13} + 2\tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \\
&- \left(5\tilde{\delta}_{15} + 5\tilde{\delta}_{16} + 4\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^2 - 2\tilde{\gamma}_1 a_i^4 \beta_B^{(i-1,i-1)} \\
&+ \tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \beta_B^{(i-1,i-1)} - \tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ 2(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} \\
&+ 2\tilde{\alpha} a_i^2 \gamma_{2,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \gamma_{3,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \nu^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - \tilde{\beta} a_i \left(\alpha_B^{(i-1,i-1)} \right)^2,
\end{aligned} \tag{12}$$

$$\begin{aligned}
\delta_{3,A}^{(i,i-1)} &= \delta_{3,B}^{(i-1,i-1)} + \tilde{\delta}_3 a_i^5 \left(\sigma^{(i-1)} \right)^2 - \left(\tilde{\delta}_5 - \tilde{\delta}_7 + \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \\
&+ \left(\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + \tilde{\delta}_{13} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 + \left(\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \\
&+ \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^2 + \tilde{\gamma}_1 a_i^4 \beta_B^{(i-1,i-1)} + (\tilde{\gamma}_2 - 2\tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ (\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \beta_B^{(i-1,i-1)} - (2\tilde{\gamma}_4 - \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - (\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
&- \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \gamma_{2,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} \\
&+ \tilde{\beta} a_i \nu^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \left(\alpha_B^{(i-1,i-1)} \right)^2,
\end{aligned} \tag{13}$$

$$\begin{aligned}
\delta_{4,A}^{(i,i-1)} &= \delta_{4,B}^{(i-1,i-1)} + \tilde{\delta}_4 a_i^5 \left(\sigma^{(i-1)} \right)^2 + \left(\tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \\
&- \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 - \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \\
&- \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^2 + 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&- 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - \tilde{\beta} a_i \left(\alpha_B^{(i-1,i-1)} \right)^2,
\end{aligned} \tag{14}$$

$$\begin{aligned}
\delta_{5,A}^{(i,i-1)} &= \delta_{5,B}^{(i-1,i-1)} + \tilde{\delta}_5 a_i^4 \left(\sigma^{(i-1)} \right)^3 - \left(\tilde{\delta}_{10} + 3\tilde{\delta}_{11} + 3\tilde{\delta}_{12} + \tilde{\delta}_{13} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
&\quad - \left(5\tilde{\delta}_{15} + 5\tilde{\delta}_{16} + 3\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 + \tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&\quad - (\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - (2\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&\quad - 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 3\tilde{\alpha} a_i^2 \gamma_{4,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \gamma_{5,B}^{(i-1,i-1)} \\
&\quad + 3\tilde{\beta} a_i \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{15}$$

$$\begin{aligned}
\delta_{6,A}^{(i,i-1)} &= \delta_{6,B}^{(i-1,i-1)} + \tilde{\delta}_6 a_i^4 \left(\sigma^{(i-1)} \right)^3 - \left(\tilde{\delta}_{10} - \tilde{\delta}_{11} - 3\tilde{\delta}_{12} + \tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
&\quad + \left(\tilde{\delta}_{15} + 3\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 + \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 - 3\tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&\quad + (3\tilde{\gamma}_4 + 2\tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + (6\tilde{\gamma}_4 + 3\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&\quad + 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} + 3\tilde{\alpha} a_i^2 \gamma_{4,B}^{(i-1,i-1)} \\
&\quad + \tilde{\beta} a_i \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{16}$$

$$\begin{aligned}
\delta_{7,A}^{(i,i-1)} &= \delta_{7,B}^{(i-1,i-1)} + \tilde{\delta}_7 a_i^4 \left(\sigma^{(i-1)} \right)^3 - \left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
&\quad - \left(2\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 - 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 \\
&\quad + (\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 3\tilde{\gamma}_4 a_i^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - 2(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&\quad - 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \gamma_{4,B}^{(i-1,i-1)} \\
&\quad + \tilde{\beta} a_i \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{17}$$

$$\begin{aligned}
\delta_{8,A}^{(i,i-1)} &= \delta_{8,B}^{(i-1,i-1)} + \tilde{\delta}_8 a_i^4 \left(\sigma^{(i-1)} \right)^3 - \left(\tilde{\delta}_{13} + 2\tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 - \left(\tilde{\delta}_{16} + 2\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
&\quad - 3\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 - 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \\
&\quad - 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{18}$$

$$\begin{aligned}
\delta_{9,A}^{(i,i-1)} &= \delta_{9,B}^{(i-1,i-1)} + \tilde{\delta}_9 a_i^4 \left(\sigma^{(i-1)} \right)^3 - \left(2\tilde{\delta}_{12} + \tilde{\delta}_{13} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 - \left(2\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
&\quad - \left(2\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 - \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 - 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&\quad + 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
&\quad - 2\tilde{\beta} a_i \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{19}$$

$$\begin{aligned}
\delta_{10,A}^{(i,i-1)} &= \delta_{10,B}^{(i-1,i-1)} + \tilde{\delta}_{10} a_i^3 \left(\sigma^{(i-1)} \right)^4 + \left(\tilde{\delta}_{15} - \tilde{\delta}_{16} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 + \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
&\quad - (2\tilde{\gamma}_4 - \tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \gamma_{6,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
&\quad + \tilde{\beta} a_i \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - \tilde{\beta} a_i \left(\beta_B^{(i-1,i-1)} \right)^2,
\end{aligned} \tag{20}$$

$$\begin{aligned}
\delta_{11,A}^{(i,i-1)} &= \delta_{11,B}^{(i-1,i-1)} + \tilde{\delta}_{11} a_i^3 \left(\sigma^{(i-1)} \right)^4 + \left(3\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 + 3\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
&\quad + \tilde{\gamma}_4 a_i^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + \tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 2\tilde{\alpha} a_i^2 \gamma_{6,B}^{(i-1,i-1)} \\
&\quad + \tilde{\beta} a_i \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{21}$$

$$\begin{aligned}
\delta_{12,A}^{(i,i-1)} &= \delta_{12,B}^{(i-1,i-1)} + \tilde{\delta}_{12} a_i^3 \left(\sigma^{(i-1)} \right)^4 + \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
&\quad + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
&\quad - 2\tilde{\beta} a_i \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{22}$$

$$\begin{aligned}
\delta_{13,A}^{(i,i-1)} &= \delta_{13,B}^{(i-1,i-1)} + \tilde{\delta}_{13} a_i^3 \left(\sigma^{(i-1)} \right)^4 + 2\tilde{\delta}_{16} a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 - \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
&\quad - 4\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
&\quad - 2\tilde{\beta} a_i \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \left(\beta_B^{(i-1,i-1)} \right)^2,
\end{aligned} \tag{23}$$

$$\begin{aligned}
\delta_{14,A}^{(i,i-1)} &= \delta_{14,B}^{(i-1,i-1)} + \tilde{\delta}_{14} a_i^3 \left(\sigma^{(i-1)} \right)^4 + 2\tilde{\delta}_{17} a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 + 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
&\quad + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
&\quad - \tilde{\beta} a_i \left(\beta_B^{(i-1,i-1)} \right)^2,
\end{aligned} \tag{24}$$

$$\begin{aligned}
\delta_{15,A}^{(i,i-1)} &= \delta_{15,B}^{(i-1,i-1)} + \tilde{\delta}_{15} a_i^2 \left(\sigma^{(i-1)} \right)^5 + \tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^5 + \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} \\
&\quad + \tilde{\beta} a_i \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{25}$$

$$\delta_{16,A}^{(i,i-1)} = \delta_{16,B}^{(i-1,i-1)} + \tilde{\delta}_{16} a_i^2 \left(\sigma^{(i-1)} \right)^5 - \tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^5 - 2\tilde{\beta} a_i \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)}, \tag{26}$$

$$\delta_{17,A}^{(i,i-1)} = \delta_{17,B}^{(i-1,i-1)} + \tilde{\delta}_{17} a_i^2 \left(\sigma^{(i-1)} \right)^5 + 5\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^5 - 2\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)}, \tag{27}$$

$$\delta_{18,A}^{(i,i-1)} = \delta_{18,B}^{(i-1,i-1)} + \tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^6. \tag{28}$$

8th order

$$\begin{aligned}
\epsilon_{1,A}^{(i,i-1)} = & \epsilon_{1,B}^{(i-1,i-1)} + \tilde{\epsilon}_1 a_i^8 \sigma^{(i-1)} + (\tilde{\epsilon}_2 + \tilde{\epsilon}_3 + \tilde{\epsilon}_4 + \tilde{\epsilon}_5) a_i^7 \nu^{(i-1)} \sigma^{(i-1)} \\
& + (\tilde{\epsilon}_6 + \tilde{\epsilon}_7 + \tilde{\epsilon}_8 + \tilde{\epsilon}_9 + \tilde{\epsilon}_{10} + \tilde{\epsilon}_{11} + \tilde{\epsilon}_{12} + \tilde{\epsilon}_{13} + \tilde{\epsilon}_{14}) a_i^6 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \\
& + (\tilde{\epsilon}_{15} + \tilde{\epsilon}_{16} + \tilde{\epsilon}_{17} + \tilde{\epsilon}_{18} + \tilde{\epsilon}_{19} + \tilde{\epsilon}_{20} + \tilde{\epsilon}_{21} + \tilde{\epsilon}_{22} + \tilde{\epsilon}_{23} + \tilde{\epsilon}_{24} + \tilde{\epsilon}_{25} + \tilde{\epsilon}_{26} + \tilde{\epsilon}_{27} + \tilde{\epsilon}_{28}) a_i^5 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \\
& - (\tilde{\epsilon}_{29} + \tilde{\epsilon}_{30} + \tilde{\epsilon}_{31} + \tilde{\epsilon}_{32} + \tilde{\epsilon}_{33} + \tilde{\epsilon}_{34} + \tilde{\epsilon}_{35} + \tilde{\epsilon}_{36} + \tilde{\epsilon}_{37} + \tilde{\epsilon}_{38} + \tilde{\epsilon}_{39} + \tilde{\epsilon}_{40} + \tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \\
& - (\tilde{\epsilon}_{43} + \tilde{\epsilon}_{44} + \tilde{\epsilon}_{45} + \tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + \tilde{\epsilon}_{49} + \tilde{\epsilon}_{50} + \tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^5 \sigma^{(i-1)} - (\tilde{\epsilon}_{52} + \tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^6 \sigma^{(i-1)} \\
& - \tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^7 \sigma^{(i-1)} + \tilde{\delta}_1 a_i^6 \alpha_B^{(i-1,i-1)} + (\tilde{\delta}_2 + \tilde{\delta}_3 + \tilde{\delta}_4) a_i^5 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \\
& + (\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9) a_i^4 \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - (\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14}) a_i^3 \left(\nu^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - (\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17}) a_i^2 \left(\nu^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^5 \alpha_B^{(i-1,i-1)} + \tilde{\gamma}_1 a_i^4 \gamma_{1,B}^{(i-1,i-1)} \\
& + (\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - (\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
& + \tilde{\alpha} a_i^2 \delta_{1,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{1,B}^{(i-1,i-1)},
\end{aligned} \tag{29}$$

$$\begin{aligned}
\epsilon_{2,A}^{(i-1,i-1)} &= \epsilon_{2,B}^{(i-1,i-1)} + \tilde{\epsilon}_{2,A} a_i^7 \left(\sigma^{(i-1)} \right)^2 + (4\tilde{\epsilon}_6 + 3\tilde{\epsilon}_7 + 5\tilde{\epsilon}_8 + 3\tilde{\epsilon}_9 + \tilde{\epsilon}_{10} + 3\tilde{\epsilon}_{12} + \tilde{\epsilon}_{13}) a_i^6 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \\
&+ (10\tilde{\epsilon}_{15} + 8\tilde{\epsilon}_{16} + 6\tilde{\epsilon}_{17} + 8\tilde{\epsilon}_{18} + 6\tilde{\epsilon}_{19} + 4\tilde{\epsilon}_{20} + 3\tilde{\epsilon}_{21} + 5\tilde{\epsilon}_{22} + 8\tilde{\epsilon}_{23} + 6\tilde{\epsilon}_{24} + 4\tilde{\epsilon}_{25} + 3\tilde{\epsilon}_{26} + 5\tilde{\epsilon}_{27} + \tilde{\epsilon}_{28}) a_i^5 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \\
&- (13\tilde{\epsilon}_{29} + 9\tilde{\epsilon}_{30} + 11\tilde{\epsilon}_{31} + 10\tilde{\epsilon}_{32} + 8\tilde{\epsilon}_{33} + 6\tilde{\epsilon}_{34} + 9\tilde{\epsilon}_{35} + 11\tilde{\epsilon}_{36} + 10\tilde{\epsilon}_{37} + 8\tilde{\epsilon}_{38} + 6\tilde{\epsilon}_{39} + 8\tilde{\epsilon}_{40} + 6\tilde{\epsilon}_{41} + 4\tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \\
&- (14\tilde{\epsilon}_{43} + 13\tilde{\epsilon}_{44} + 11\tilde{\epsilon}_{45} + 14\tilde{\epsilon}_{46} + 13\tilde{\epsilon}_{47} + 11\tilde{\epsilon}_{48} + 9\tilde{\epsilon}_{49} + 11\tilde{\epsilon}_{50} + 10\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^2 \\
&- (14\tilde{\epsilon}_{52} + 14\tilde{\epsilon}_{53} + 14\tilde{\epsilon}_{54} + 13\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^2 - 14\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^6 \left(\sigma^{(i-1)} \right)^2 - 5\tilde{\delta}_1 a_i^6 \beta_B^{(i-1,i-1)} \\
&+ \left(\tilde{\delta}_2 + \tilde{\delta}_3 \right) a_i^5 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 5 \left(\tilde{\delta}_2 + \tilde{\delta}_3 + \tilde{\delta}_4 \right) a_i^5 \nu^{(i-1)} \beta_B^{(i-1,i-1)} + \left(2\tilde{\delta}_5 + 2\tilde{\delta}_6 + 2\tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&- 5 \left(\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \left(\nu^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - \left(3\tilde{\delta}_{10} + 3\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + 2\tilde{\delta}_{13} + 2\tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ 5 \left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} - \left(4\tilde{\delta}_{15} + 3\tilde{\delta}_{16} + 3\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
&+ 5 \left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} - 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^5 \beta_B^{(i-1,i-1)} \\
&+ 5\tilde{\gamma}_1 a_i^4 \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_1 a_i^4 \gamma_{3,B}^{(i-1,i-1)} - (\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} + 5(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
&+ 3(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + 2(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 5(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
&- 3(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 5\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{2,B}^{(i-1,i-1)} \\
&- 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{3,B}^{(i-1,i-1)} + \tilde{\gamma}_2 a_i^3 \left(\alpha_B^{(i-1,i-1)} \right)^2 - (2\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
&+ 3\tilde{\alpha} a_i^2 \delta_{2,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{3,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{2,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{3,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned} \tag{30}$$

$$\begin{aligned}
\epsilon_{3,A}^{(i,i-1)} = & \epsilon_{3,B}^{(i-1,i-1)} + \tilde{\epsilon}_3 a_i^7 \left(\sigma^{(i-1)} \right)^2 - (3\tilde{\epsilon}_6 + 2\tilde{\epsilon}_7 + 5\tilde{\epsilon}_8 + 3\tilde{\epsilon}_9 - \tilde{\epsilon}_{11} + 3\tilde{\epsilon}_{12} - \tilde{\epsilon}_{14}) a_i^6 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \\
& - (11\tilde{\epsilon}_{15} + 8\tilde{\epsilon}_{16} + 5\tilde{\epsilon}_{17} + 9\tilde{\epsilon}_{18} + 6\tilde{\epsilon}_{19} + 3\tilde{\epsilon}_{20} + 2\tilde{\epsilon}_{21} + 5\tilde{\epsilon}_{22} + 9\tilde{\epsilon}_{23} + 6\tilde{\epsilon}_{24} + 3\tilde{\epsilon}_{25} + 2\tilde{\epsilon}_{26} + 5\tilde{\epsilon}_{27}) a_i^5 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \\
& + (14\tilde{\epsilon}_{29} + 9\tilde{\epsilon}_{30} + 12\tilde{\epsilon}_{31} + 11\tilde{\epsilon}_{32} + 8\tilde{\epsilon}_{33} + 5\tilde{\epsilon}_{34} + 9\tilde{\epsilon}_{35} + 12\tilde{\epsilon}_{36} + 11\tilde{\epsilon}_{37} + 8\tilde{\epsilon}_{38} + 5\tilde{\epsilon}_{39} + 9\tilde{\epsilon}_{40} + 6\tilde{\epsilon}_{41} + 3\tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \\
& + (15\tilde{\epsilon}_{43} + 14\tilde{\epsilon}_{44} + 11\tilde{\epsilon}_{45} + 15\tilde{\epsilon}_{46} + 14\tilde{\epsilon}_{47} + 11\tilde{\epsilon}_{48} + 9\tilde{\epsilon}_{49} + 12\tilde{\epsilon}_{50} + 11\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^2 \\
& + (14\tilde{\epsilon}_{52} + 14\tilde{\epsilon}_{53} + 15\tilde{\epsilon}_{54} + 14\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^2 + 14\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^6 \left(\sigma^{(i-1)} \right)^2 + 6\tilde{\delta}_1 a_i^6 \beta_B^{(i-1,i-1)} \\
& - \left(\tilde{\delta}_2 + 2\tilde{\delta}_3 - \tilde{\delta}_4 \right) a_i^5 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + 6 \left(\tilde{\delta}_2 + \tilde{\delta}_3 + \tilde{\delta}_4 \right) a_i^5 \nu^{(i-1)} \beta_B^{(i-1,i-1)} - \left(\tilde{\delta}_5 + 2\tilde{\delta}_6 + 3\tilde{\delta}_7 - \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
& + 6 \left(\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \left(\nu^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + \left(3\tilde{\delta}_{10} + 2\tilde{\delta}_{11} - \tilde{\delta}_{12} + \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
& - 6 \left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} + \left(3\tilde{\delta}_{15} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
& - 6 \left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 6\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^5 \beta_B^{(i-1,i-1)} \\
& - 6\tilde{\gamma}_1 a_i^4 \gamma_{2,B}^{(i-1,i-1)} - 3\tilde{\gamma}_1 a_i^4 \gamma_{3,B}^{(i-1,i-1)} + (2\tilde{\gamma}_2 + 3\tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 6(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
& - 3(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - (4\tilde{\gamma}_4 + 5\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} + 6(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
& + 3(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 7\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{2,B}^{(i-1,i-1)} \\
& + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{3,B}^{(i-1,i-1)} - (2\tilde{\gamma}_2 - \tilde{\gamma}_3) a_i^3 \left(\alpha_B^{(i-1,i-1)} \right)^2 + (4\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& - 3\tilde{\alpha}_a a_i^2 \delta_{2,B}^{(i-1,i-1)} + \tilde{\alpha}_a^2 \delta_{4,B}^{(i-1,i-1)} + 3\tilde{\beta}_a a_i \nu^{(i-1)} \delta_{2,B}^{(i-1,i-1)} - \tilde{\beta}_a a_i \nu^{(i-1)} \delta_{4,B}^{(i-1,i-1)} - 4\tilde{\beta}_a a_i \alpha_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned} \tag{31}$$

$$\begin{aligned}
\epsilon_{4,A}^{(i,i-1)} = & \epsilon_{4,B}^{(i-1,i-1)} + \tilde{\epsilon}_4 a_i^7 \left(\sigma^{(i-1)} \right)^2 + (\tilde{\epsilon}_6 + \tilde{\epsilon}_7 + 2\tilde{\epsilon}_8 + 2\tilde{\epsilon}_9 + \tilde{\epsilon}_{10} + \tilde{\epsilon}_{11} + \tilde{\epsilon}_{12}) a_i^6 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \\
& + (4\tilde{\epsilon}_{15} + 3\tilde{\epsilon}_{16} + 2\tilde{\epsilon}_{17} + 4\tilde{\epsilon}_{18} + 3\tilde{\epsilon}_{19} + 2\tilde{\epsilon}_{20} + 2\tilde{\epsilon}_{21} + 3\tilde{\epsilon}_{22} + 3\tilde{\epsilon}_{23} + 2\tilde{\epsilon}_{24} + \tilde{\epsilon}_{25} + \tilde{\epsilon}_{26} + 2\tilde{\epsilon}_{27} + \tilde{\epsilon}_{28}) a_i^5 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \\
& - (5\tilde{\epsilon}_{29} + 4\tilde{\epsilon}_{30} + 5\tilde{\epsilon}_{31} + 5\tilde{\epsilon}_{32} + 4\tilde{\epsilon}_{33} + 3\tilde{\epsilon}_{34} + 3\tilde{\epsilon}_{35} + 4\tilde{\epsilon}_{36} + 4\tilde{\epsilon}_{37} + 3\tilde{\epsilon}_{38} + 2\tilde{\epsilon}_{39} + 4\tilde{\epsilon}_{40} + 3\tilde{\epsilon}_{41} + 2\tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \\
& - (6\tilde{\epsilon}_{43} + 6\tilde{\epsilon}_{44} + 5\tilde{\epsilon}_{45} + 5\tilde{\epsilon}_{46} + 5\tilde{\epsilon}_{47} + 4\tilde{\epsilon}_{48} + 4\tilde{\epsilon}_{49} + 5\tilde{\epsilon}_{50} + 5\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^2 \\
& - (6\tilde{\epsilon}_{52} + 5\tilde{\epsilon}_{53} + 6\tilde{\epsilon}_{54} + 6\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^2 - 6\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^6 \left(\sigma^{(i-1)} \right)^2 - 2\tilde{\delta}_1 a_i^6 \beta_B^{(i-1,i-1)} \\
& + \left(\tilde{\delta}_2 + 2\tilde{\delta}_3 - 2\tilde{\delta}_4 \right) a_i^5 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2 \left(\tilde{\delta}_2 + \tilde{\delta}_3 + \tilde{\delta}_4 \right) a_i^5 \nu^{(i-1)} \beta_B^{(i-1,i-1)} + \left(\tilde{\delta}_5 + 2\tilde{\delta}_6 + 3\tilde{\delta}_7 - \tilde{\delta}_8 - 2\tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
& - 2 \left(\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \left(\nu^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - \left(3\tilde{\delta}_{10} + 2\tilde{\delta}_{11} - 2\tilde{\delta}_{12} - \tilde{\delta}_{13} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
& + 2 \left(\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} - \left(3\tilde{\delta}_{15} - \tilde{\delta}_{16} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
& + 2 \left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^5 \beta_B^{(i-1,i-1)} \\
& + 2\tilde{\gamma}_1 a_i^4 \gamma_{2,B}^{(i-1,i-1)} + \tilde{\gamma}_1 a_i^4 \gamma_{3,B}^{(i-1,i-1)} - 3\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} + 2 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
& + 3\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 2 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{2,B}^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{3,B}^{(i-1,i-1)} + \tilde{\gamma}_2 a_i^3 \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& - 2\tilde{\gamma}_3 a_i^3 \left(\alpha_B^{(i-1,i-1)} \right)^2 - (2\tilde{\gamma}_4 - \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 + \tilde{\alpha} a_i^2 \delta_{2,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{1,B}^{(i-1,i-1)} \\
& - \tilde{\beta} a_i \nu^{(i-1)} \delta_{2,B}^{(i-1,i-1)} + 5\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned} \tag{32}$$

$$\begin{aligned}
\epsilon_{5,A}^{(i,i-1)} = & \epsilon_{5,B}^{(i-1,i-1)} + \tilde{\epsilon}_5 a_i^7 \left(\sigma^{(i-1)} \right)^2 + (\tilde{\epsilon}_{12} + \tilde{\epsilon}_{13} + \tilde{\epsilon}_{14}) a_i^6 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 + (\tilde{\epsilon}_{23} + \tilde{\epsilon}_{24} + \tilde{\epsilon}_{25} + \tilde{\epsilon}_{26} + \tilde{\epsilon}_{27} + \tilde{\epsilon}_{28}) a_i^5 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \\
& - (\tilde{\epsilon}_{35} + \tilde{\epsilon}_{36} + \tilde{\epsilon}_{37} + \tilde{\epsilon}_{38} + \tilde{\epsilon}_{39} + \tilde{\epsilon}_{40} + \tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 - (\tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + \tilde{\epsilon}_{49} + \tilde{\epsilon}_{50} + \tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^2 \\
& - (\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^2 - \tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^6 \left(\sigma^{(i-1)} \right)^2 + 2\tilde{\delta}_4 a_i^5 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
& + 2 \left(\tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2 \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \\
& - 2 \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} - 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} + 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} \\
& - 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{1,B}^{(i-1,i-1)} + \tilde{\gamma}_3 a_i^3 \left(\alpha_B^{(i-1,i-1)} \right)^2 - \tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{1,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned} \tag{33}$$

$$\begin{aligned}
\epsilon_{6,A}^{(i,i-1)} = & \epsilon_{6,B}^{(i-1,i-1)} + \tilde{\epsilon}_6 a_i^6 \left(\sigma^{(i-1)} \right)^3 + (10\tilde{\epsilon}_{15} + 4\tilde{\epsilon}_{16} + \tilde{\epsilon}_{17} + 4\tilde{\epsilon}_{18} + \tilde{\epsilon}_{19} + \tilde{\epsilon}_{20} + 4\tilde{\epsilon}_{23} + \tilde{\epsilon}_{24} + \tilde{\epsilon}_{25}) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
& - (18\tilde{\epsilon}_{29} + 6\tilde{\epsilon}_{30} + 12\tilde{\epsilon}_{31} + 10\tilde{\epsilon}_{32} + 4\tilde{\epsilon}_{33} + \tilde{\epsilon}_{34} + 6\tilde{\epsilon}_{35} + 12\tilde{\epsilon}_{36} + 10\tilde{\epsilon}_{37} + 4\tilde{\epsilon}_{38} + \tilde{\epsilon}_{39} + 4\tilde{\epsilon}_{40} + \tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
& - (21\tilde{\epsilon}_{43} + 18\tilde{\epsilon}_{44} + 12\tilde{\epsilon}_{45} + 21\tilde{\epsilon}_{46} + 18\tilde{\epsilon}_{47} + 12\tilde{\epsilon}_{48} + 6\tilde{\epsilon}_{49} + 12\tilde{\epsilon}_{50} + 10\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 \\
& - (21\tilde{\epsilon}_{52} + 21\tilde{\epsilon}_{53} + 21\tilde{\epsilon}_{54} + 18\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^3 - 21\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^3 + \tilde{\delta}_2 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + \left(2\tilde{\delta}_5 + \tilde{\delta}_6 + \tilde{\delta}_7 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + \left(3\tilde{\delta}_5 + 2\tilde{\delta}_6 + \tilde{\delta}_7 + \tilde{\delta}_8 + 2\tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - \left(4\tilde{\delta}_{10} + 6\tilde{\delta}_{11} + 4\tilde{\delta}_{12} + 2\tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \left(3\tilde{\delta}_{10} + 4\tilde{\delta}_{11} + 4\tilde{\delta}_{12} + 3\tilde{\delta}_{13} + 2\tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - \left(11\tilde{\delta}_{15} + 8\tilde{\delta}_{16} + 6\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \left(5\tilde{\delta}_{15} + 5\tilde{\delta}_{16} + 4\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - 11\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 10\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} - 4\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
& - \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - 10 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 4 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& + 2 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 2 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + 10 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
& + 4 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
& + 10\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} - \tilde{\gamma}_2 a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - \tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& + (2\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& + 4\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{7,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} \\
& - \tilde{\beta} a_i \nu^{(i-1)} \delta_{7,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{34}$$

$$\begin{aligned}
\epsilon_{7,A}^{(i,i-1)} = & \epsilon_{7,B}^{(i-1,i-1)} + \tilde{\epsilon}_7 a_i^6 \left(\sigma^{(i-1)} \right)^3 - (30\tilde{\epsilon}_{15} + 13\tilde{\epsilon}_{16} + 2\tilde{\epsilon}_{17} + 14\tilde{\epsilon}_{18} + 4\tilde{\epsilon}_{19} - \tilde{\epsilon}_{21} + 14\tilde{\epsilon}_{23} + 4\tilde{\epsilon}_{24} - \tilde{\epsilon}_{26}) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
& + (57\tilde{\epsilon}_{29} + 18\tilde{\epsilon}_{30} + 38\tilde{\epsilon}_{31} + 30\tilde{\epsilon}_{32} + 13\tilde{\epsilon}_{33} + 2\tilde{\epsilon}_{34} + 18\tilde{\epsilon}_{35} + 38\tilde{\epsilon}_{36} + 30\tilde{\epsilon}_{37} + 13\tilde{\epsilon}_{38} + 2\tilde{\epsilon}_{39} + 14\tilde{\epsilon}_{40} + 4\tilde{\epsilon}_{41}) a_i^4 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
& + (66\tilde{\epsilon}_{43} + 57\tilde{\epsilon}_{44} + 36\tilde{\epsilon}_{45} + 66\tilde{\epsilon}_{46} + 57\tilde{\epsilon}_{47} + 36\tilde{\epsilon}_{48} + 18\tilde{\epsilon}_{49} + 38\tilde{\epsilon}_{50} + 30\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 \\
& + (63\tilde{\epsilon}_{52} + 63\tilde{\epsilon}_{53} + 66\tilde{\epsilon}_{54} + 57\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^3 + 63\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^3 + 2\tilde{\delta}_3 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - (4\tilde{\delta}_5 + 4\tilde{\delta}_6 + 2\tilde{\delta}_7 - \tilde{\delta}_8) a_i^4 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - 2 \left(\tilde{\delta}_5 - \tilde{\delta}_7 + \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + (10\tilde{\delta}_{10} + 10\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + 3\tilde{\delta}_{13} + 2\tilde{\delta}_{14}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + 2 \left(\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + \tilde{\delta}_{13} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + (20\tilde{\delta}_{15} + 9\tilde{\delta}_{16} + 8\tilde{\delta}_{17}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + 2 \left(\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + 17\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} + 30\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} + 14\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
& + 4 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + (2\tilde{\gamma}_2 + 3\tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + 30 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 14 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& - 2 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} - 8 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - (4\tilde{\gamma}_4 + 5\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
& - 30 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} - 14 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} \\
& - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 7\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - 30\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} - 14\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
& + 6\tilde{\gamma}_2 a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 6 \left(2\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 - 14\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} - 4\tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{8,B}^{(i-1,i-1)} \\
& + 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} + 14\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} \\
& + 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{8,B}^{(i-1,i-1)},
\end{aligned} \tag{35}$$

$$\begin{aligned}
\epsilon_{8,A}^{(i,i-1)} = & \epsilon_{8,B}^{(i-1,i-1)} + \tilde{\epsilon}_8 a_i^6 \left(\sigma^{(i-1)} \right)^3 + (17\tilde{\epsilon}_{15} + 10\tilde{\epsilon}_{16} + 4\tilde{\epsilon}_{17} + 9\tilde{\epsilon}_{18} + 4\tilde{\epsilon}_{19} + \tilde{\epsilon}_{22} + 9\tilde{\epsilon}_{23} + 4\tilde{\epsilon}_{24} + \tilde{\epsilon}_{27}) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
& - (35\tilde{\epsilon}_{29} + 13\tilde{\epsilon}_{30} + 23\tilde{\epsilon}_{31} + 17\tilde{\epsilon}_{32} + 10\tilde{\epsilon}_{33} + 4\tilde{\epsilon}_{34} + 13\tilde{\epsilon}_{35} + 23\tilde{\epsilon}_{36} + 17\tilde{\epsilon}_{37} + 10\tilde{\epsilon}_{38} + 4\tilde{\epsilon}_{39} + 9\tilde{\epsilon}_{40} + 4\tilde{\epsilon}_{41}) a_i^4 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
& - (41\tilde{\epsilon}_{43} + 35\tilde{\epsilon}_{44} + 24\tilde{\epsilon}_{45} + 41\tilde{\epsilon}_{46} + 35\tilde{\epsilon}_{47} + 24\tilde{\epsilon}_{48} + 13\tilde{\epsilon}_{49} + 23\tilde{\epsilon}_{50} + 17\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 \\
& - (42\tilde{\epsilon}_{52} + 42\tilde{\epsilon}_{53} + 41\tilde{\epsilon}_{54} + 35\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^3 - 42\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^3 - (2\tilde{\delta}_2 + \tilde{\delta}_3) a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + (\tilde{\delta}_5 + 2\tilde{\delta}_6 + \tilde{\delta}_9) a_i^4 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - (5\tilde{\delta}_5 + 4\tilde{\delta}_6 + 3\tilde{\delta}_7 + 2\tilde{\delta}_8 + 3\tilde{\delta}_9) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - (3\tilde{\delta}_{10} + \tilde{\delta}_{11} - \tilde{\delta}_{12} + 2\tilde{\delta}_{13} + 2\tilde{\delta}_{14}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + (6\tilde{\delta}_{10} + 7\tilde{\delta}_{11} + 6\tilde{\delta}_{12} + 5\tilde{\delta}_{13} + 4\tilde{\delta}_{14}) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - (3\tilde{\delta}_{15} + \tilde{\delta}_{16} + 2\tilde{\delta}_{17}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + (9\tilde{\delta}_{15} + 8\tilde{\delta}_{16} + 7\tilde{\delta}_{17}) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + 9\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 15\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} - 9\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
& - (2\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 15 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 9 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& + (\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} + (4\tilde{\gamma}_4 + 3\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 15 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
& + 9 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} + 5\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
& + 15\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} + 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} - (3\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + (2\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& + (6\tilde{\gamma}_4 + 4\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 + 7\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& + 9\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} + 4\tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{9,B}^{(i-1,i-1)} - 9\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} \\
& - \tilde{\beta} a_i \nu^{(i-1)} \delta_{9,B}^{(i-1,i-1)} - \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)},
\end{aligned} \tag{36}$$

$$\begin{aligned}
\epsilon_{9,A}^{(i,i-1)} = & \epsilon_{9,B}^{(i-1,i-1)} + \tilde{\epsilon}_9 a_i^6 \left(\sigma^{(i-1)} \right)^3 - (30\tilde{\epsilon}_{15} + 16\tilde{\epsilon}_{16} + 6\tilde{\epsilon}_{17} + 13\tilde{\epsilon}_{18} + 4\tilde{\epsilon}_{19} - \tilde{\epsilon}_{20} - \tilde{\epsilon}_{21} - 2\tilde{\epsilon}_{22} + 16\tilde{\epsilon}_{23} + 6\tilde{\epsilon}_{24}) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
& + (62\tilde{\epsilon}_{29} + 19\tilde{\epsilon}_{30} + 38\tilde{\epsilon}_{31} + 26\tilde{\epsilon}_{32} + 13\tilde{\epsilon}_{33} + 4\tilde{\epsilon}_{34} + 22\tilde{\epsilon}_{35} + 42\tilde{\epsilon}_{36} + 30\tilde{\epsilon}_{37} + 16\tilde{\epsilon}_{38} + 6\tilde{\epsilon}_{39} + 13\tilde{\epsilon}_{40} + 4\tilde{\epsilon}_{41} - \tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
& + (69\tilde{\epsilon}_{43} + 57\tilde{\epsilon}_{44} + 38\tilde{\epsilon}_{45} + 74\tilde{\epsilon}_{46} + 62\tilde{\epsilon}_{47} + 42\tilde{\epsilon}_{48} + 19\tilde{\epsilon}_{49} + 38\tilde{\epsilon}_{50} + 26\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 \\
& + (69\tilde{\epsilon}_{52} + 74\tilde{\epsilon}_{53} + 69\tilde{\epsilon}_{54} + 57\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^3 + 69\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^3 - 2\tilde{\delta}_3 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - (4\tilde{\delta}_5 + 4\tilde{\delta}_6 - 2\tilde{\delta}_7 + 2\tilde{\delta}_8 + 4\tilde{\delta}_9) a_i^4 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + 2 \left(\tilde{\delta}_5 - \tilde{\delta}_7 + \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + (6\tilde{\delta}_{10} + 6\tilde{\delta}_{11} + 6\tilde{\delta}_{12} + 10\tilde{\delta}_{13} + 8\tilde{\delta}_{14}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - 2 \left(\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + \tilde{\delta}_{13} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + (12\tilde{\delta}_{15} + 16\tilde{\delta}_{16} + 14\tilde{\delta}_{17}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - 2 \left(\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + 22\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} + 30\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} + 16\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
& + 6\tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 3\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + 30 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 16 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& - 6 \left(2\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - 30 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
& - 16 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
& - 30\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} - 16\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} + 4 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 7\tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& - 8 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& - 16\tilde{\alpha}_i^2 \delta_{5,B}^{(i-1,i-1)} - 6\tilde{\alpha}_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{2,B}^{(i-1,i-1)} + 16\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} + 6\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} \\
& + 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 5\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)}, \tag{37}
\end{aligned}$$

$$\begin{aligned}
\epsilon_{10,A}^{(i,i-1)} = & \epsilon_{10,B}^{(i-1,i-1)} + \tilde{\epsilon}_{10} a_i^6 \left(\sigma^{(i-1)} \right)^3 + (45\tilde{\epsilon}_{15} + 22\tilde{\epsilon}_{16} + 7\tilde{\epsilon}_{17} + 21\tilde{\epsilon}_{18} + 7\tilde{\epsilon}_{19} + \tilde{\epsilon}_{20} - \tilde{\epsilon}_{22} + 22\tilde{\epsilon}_{23} + 7\tilde{\epsilon}_{24} + \tilde{\epsilon}_{28}) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
& - (89\tilde{\epsilon}_{29} + 29\tilde{\epsilon}_{30} + 58\tilde{\epsilon}_{31} + 43\tilde{\epsilon}_{32} + 21\tilde{\epsilon}_{33} + 7\tilde{\epsilon}_{34} + 29\tilde{\epsilon}_{35} + 59\tilde{\epsilon}_{36} + 45\tilde{\epsilon}_{37} + 22\tilde{\epsilon}_{38} + 7\tilde{\epsilon}_{39} + 21\tilde{\epsilon}_{40} + 7\tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
& - (102\tilde{\epsilon}_{43} + 87\tilde{\epsilon}_{44} + 58\tilde{\epsilon}_{45} + 103\tilde{\epsilon}_{46} + 89\tilde{\epsilon}_{47} + 59\tilde{\epsilon}_{48} + 29\tilde{\epsilon}_{49} + 58\tilde{\epsilon}_{50} + 43\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 \\
& - (102\tilde{\epsilon}_{52} + 103\tilde{\epsilon}_{53} + 102\tilde{\epsilon}_{54} + 87\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^3 - 102\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^3 - \tilde{\delta}_3 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + (8\tilde{\delta}_5 + 7\tilde{\delta}_6 - 2\tilde{\delta}_7 + 2\tilde{\delta}_9) a_i^4 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_5 - \tilde{\delta}_7 + \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - (13\tilde{\delta}_{10} + 15\tilde{\delta}_{11} + 5\tilde{\delta}_{12} + 10\tilde{\delta}_{13} + 7\tilde{\delta}_{14}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{11} + 2\tilde{\delta}_{12} + \tilde{\delta}_{13} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - (29\tilde{\delta}_{15} + 21\tilde{\delta}_{16} + 17\tilde{\delta}_{17}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - 31\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 45\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} - 22\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
& - (7\tilde{\gamma}_2 + 4\tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 45 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 22 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& + (3\tilde{\gamma}_4 + 2\tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} + (14\tilde{\gamma}_4 + 11\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 45 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
& + 22 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} + 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} + 18\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
& + 45\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} + 22\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} - 2 \left(4\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& + 2 \left(4\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 + (16\tilde{\gamma}_4 + 10\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 12\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& + 18\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 22\tilde{\alpha}_i^2 \delta_{5,B}^{(i-1,i-1)} + 7\tilde{\alpha}_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{3,B}^{(i-1,i-1)} - 22\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} \\
& - 7\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)}, \tag{38}
\end{aligned}$$

$$\begin{aligned}
\epsilon_{11,A}^{(i,i-1)} = & \epsilon_{11,B}^{(i-1,i-1)} + \tilde{\epsilon}_{11} a_i^6 \left(\sigma^{(i-1)} \right)^3 - (9\tilde{\epsilon}_{15} + 4\tilde{\epsilon}_{16} + \tilde{\epsilon}_{17} + 4\tilde{\epsilon}_{18} + \tilde{\epsilon}_{19} - \tilde{\epsilon}_{21} - \tilde{\epsilon}_{22} + 4\tilde{\epsilon}_{23} + \tilde{\epsilon}_{24}) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
& + (17\tilde{\epsilon}_{29} + 5\tilde{\epsilon}_{30} + 11\tilde{\epsilon}_{31} + 8\tilde{\epsilon}_{32} + 3\tilde{\epsilon}_{33} + 5\tilde{\epsilon}_{35} + 11\tilde{\epsilon}_{36} + 9\tilde{\epsilon}_{37} + 4\tilde{\epsilon}_{38} + \tilde{\epsilon}_{39} + 4\tilde{\epsilon}_{40} + \tilde{\epsilon}_{41}) a_i^4 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
& + (19\tilde{\epsilon}_{43} + 16\tilde{\epsilon}_{44} + 10\tilde{\epsilon}_{45} + 19\tilde{\epsilon}_{46} + 17\tilde{\epsilon}_{47} + 11\tilde{\epsilon}_{48} + 5\tilde{\epsilon}_{49} + 11\tilde{\epsilon}_{50} + 8\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 \\
& + (18\tilde{\epsilon}_{52} + 19\tilde{\epsilon}_{53} + 19\tilde{\epsilon}_{54} + 16\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^3 + 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^3 + \left(\tilde{\delta}_3 + \tilde{\delta}_4 \right) a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - \left(2\tilde{\delta}_5 + \tilde{\delta}_6 - 2\tilde{\delta}_7 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_5 - \tilde{\delta}_7 - \tilde{\delta}_8 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + \left(\tilde{\delta}_{10} + 3\tilde{\delta}_{11} + \tilde{\delta}_{12} + 2\tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_{11} + \tilde{\delta}_{12} - \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + \left(5\tilde{\delta}_{15} + 5\tilde{\delta}_{16} + 3\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_{15} + \tilde{\delta}_{16} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + 9\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} + 4\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} + (\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
& + 9(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 4(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 3\tilde{\gamma}_4 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} \\
& - 2(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 9(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} - 4(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
& - 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
& + (2\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 - (4\tilde{\gamma}_4 + 3\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 - 5\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 4\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{4,B}^{(i-1,i-1)} \\
& + 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)},
\end{aligned} \tag{39}$$

$$\begin{aligned}
\epsilon_{12,A}^{(i,i-1)} = & \epsilon_{12,B}^{(i-1,i-1)} + \tilde{\epsilon}_{12} a_i^6 \left(\sigma^{(i-1)} \right)^3 + (18\tilde{\epsilon}_{15} + 9\tilde{\epsilon}_{16} + 3\tilde{\epsilon}_{17} + 9\tilde{\epsilon}_{18} + 3\tilde{\epsilon}_{19} + 12\tilde{\epsilon}_{23} + 5\tilde{\epsilon}_{24} + \tilde{\epsilon}_{25} + \tilde{\epsilon}_{26} + 2\tilde{\epsilon}_{27} + \tilde{\epsilon}_{28}) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
& - (36\tilde{\epsilon}_{29} + 12\tilde{\epsilon}_{30} + 24\tilde{\epsilon}_{31} + 18\tilde{\epsilon}_{32} + 9\tilde{\epsilon}_{33} + 3\tilde{\epsilon}_{34} + 15\tilde{\epsilon}_{35} + 28\tilde{\epsilon}_{36} + 22\tilde{\epsilon}_{37} + 12\tilde{\epsilon}_{38} + 5\tilde{\epsilon}_{39} + 13\tilde{\epsilon}_{40} + 6\tilde{\epsilon}_{41} + 2\tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
& - (42\tilde{\epsilon}_{43} + 36\tilde{\epsilon}_{44} + 24\tilde{\epsilon}_{45} + 47\tilde{\epsilon}_{46} + 41\tilde{\epsilon}_{47} + 28\tilde{\epsilon}_{48} + 16\tilde{\epsilon}_{49} + 29\tilde{\epsilon}_{50} + 23\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 \\
& - (42\tilde{\epsilon}_{52} + 47\tilde{\epsilon}_{53} + 48\tilde{\epsilon}_{54} + 42\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^3 - 48\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^3 - \tilde{\delta}_4 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + \left(3\tilde{\delta}_5 + 3\tilde{\delta}_6 + 2\tilde{\delta}_8 + 2\tilde{\delta}_9 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - \left(6\tilde{\delta}_{10} + 6\tilde{\delta}_{11} + 4\tilde{\delta}_{12} + 7\tilde{\delta}_{13} + 7\tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - 12 \left(\tilde{\delta}_{15} + \tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - 20\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 18\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} - 9\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
& - (3\tilde{\gamma}_2 - \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} - 18(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 9(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} + 2(3\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
& + 18(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} + 9(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} \\
& + 5\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{3,B}^{(i-1,i-1)} + 18\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} + 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
& - (3\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + (3\tilde{\gamma}_4 - \tilde{\gamma}_5) a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 + (6\tilde{\gamma}_4 + 4\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& + \tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 + 7\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 9\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} + 3\tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} \\
& - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{2,B}^{(i-1,i-1)} - 9\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} \\
& - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{40}$$

$$\begin{aligned}
\epsilon_{13,A}^{(i,i-1)} = & \epsilon_{13,B}^{(i-1,i-1)} + \tilde{\epsilon}_{13} a_i^6 \left(\sigma^{(i-1)} \right)^3 - (24\tilde{\epsilon}_{15} + 12\tilde{\epsilon}_{16} + 4\tilde{\epsilon}_{17} + 12\tilde{\epsilon}_{18} + 4\tilde{\epsilon}_{19} + 13\tilde{\epsilon}_{23} + 4\tilde{\epsilon}_{24} - \tilde{\epsilon}_{25} + \tilde{\epsilon}_{27} + 2\tilde{\epsilon}_{28}) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
& + (48\tilde{\epsilon}_{29} + 16\tilde{\epsilon}_{30} + 32\tilde{\epsilon}_{31} + 24\tilde{\epsilon}_{32} + 12\tilde{\epsilon}_{33} + 4\tilde{\epsilon}_{34} + 16\tilde{\epsilon}_{35} + 33\tilde{\epsilon}_{36} + 26\tilde{\epsilon}_{37} + 13\tilde{\epsilon}_{38} + 4\tilde{\epsilon}_{39} + 16\tilde{\epsilon}_{40} + 7\tilde{\epsilon}_{41} + 2\tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
& + (56\tilde{\epsilon}_{43} + 48\tilde{\epsilon}_{44} + 32\tilde{\epsilon}_{45} + 57\tilde{\epsilon}_{46} + 50\tilde{\epsilon}_{47} + 33\tilde{\epsilon}_{48} + 19\tilde{\epsilon}_{49} + 36\tilde{\epsilon}_{50} + 29\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 \\
& + (56\tilde{\epsilon}_{52} + 57\tilde{\epsilon}_{53} + 60\tilde{\epsilon}_{54} + 53\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^3 + 60\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^3 - 2\tilde{\delta}_4 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - 2 \left(2\tilde{\delta}_5 + 2\tilde{\delta}_6 + \tilde{\delta}_8 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - 2 \left(\tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + \left(8\tilde{\delta}_{10} + 8\tilde{\delta}_{11} + 6\tilde{\delta}_{13} + 8\tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + 2 \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + \left(16\tilde{\delta}_{15} + 10\tilde{\delta}_{16} + 12\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + 2 \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + 22\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} + 24\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} + 12\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
& + 2 \left(2\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 24 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 12 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} \\
& - \left(8\tilde{\gamma}_4 + 6\tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 24 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} - 12 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
& - 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} - 10\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 24\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
& + 2 \left(2\tilde{\gamma}_2 - \tilde{\gamma}_3 \right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - \left(4\tilde{\gamma}_4 - \tilde{\gamma}_5 \right) a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 - 2 \left(4\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 12\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} - 4\tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} \\
& - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{3,B}^{(i-1,i-1)} + 12\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} + 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)},
\end{aligned} \tag{41}$$

$$\begin{aligned}
\epsilon_{14,A}^{(i,i-1)} = & \epsilon_{14,B}^{(i-1,i-1)} + \tilde{\epsilon}_{14} a_i^6 \left(\sigma^{(i-1)} \right)^3 + (6\tilde{\epsilon}_{15} + 3\tilde{\epsilon}_{16} + \tilde{\epsilon}_{17} + 3\tilde{\epsilon}_{18} + \tilde{\epsilon}_{19} + 3\tilde{\epsilon}_{23} + \tilde{\epsilon}_{24} + \tilde{\epsilon}_{26} + \tilde{\epsilon}_{27} + 3\tilde{\epsilon}_{28}) a_i^5 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \\
& - (12\tilde{\epsilon}_{29} + 4\tilde{\epsilon}_{30} + 8\tilde{\epsilon}_{31} + 6\tilde{\epsilon}_{32} + 3\tilde{\epsilon}_{33} + \tilde{\epsilon}_{34} + 4\tilde{\epsilon}_{35} + 8\tilde{\epsilon}_{36} + 7\tilde{\epsilon}_{37} + 4\tilde{\epsilon}_{38} + 2\tilde{\epsilon}_{39} + 6\tilde{\epsilon}_{40} + 4\tilde{\epsilon}_{41} + 3\tilde{\epsilon}_{42}) a_i^4 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \\
& - (14\tilde{\epsilon}_{43} + 12\tilde{\epsilon}_{44} + 8\tilde{\epsilon}_{45} + 14\tilde{\epsilon}_{46} + 13\tilde{\epsilon}_{47} + 9\tilde{\epsilon}_{48} + 7\tilde{\epsilon}_{49} + 11\tilde{\epsilon}_{50} + 10\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^3 \\
& - (14\tilde{\epsilon}_{52} + 15\tilde{\epsilon}_{53} + 17\tilde{\epsilon}_{54} + 16\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^3 - 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^5 \left(\sigma^{(i-1)} \right)^3 + \tilde{\delta}_4 a_i^5 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& + \left(\tilde{\delta}_5 + \tilde{\delta}_6 + 2\tilde{\delta}_8 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \nu^{(i-1)} \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - \left(2\tilde{\delta}_{10} + 2\tilde{\delta}_{11} + 3\tilde{\delta}_{13} + 5\tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - \left(4\tilde{\delta}_{15} + 4\tilde{\delta}_{16} + 6\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} \\
& - 10\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^4 \sigma^{(i-1)} \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_1 a_i^4 \gamma_{4,B}^{(i-1,i-1)} - 3\tilde{\gamma}_1 a_i^4 \gamma_{5,B}^{(i-1,i-1)} \\
& - \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} - 6 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 3 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} \\
& + 2 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 6 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} + 3 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
& - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{1,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{2,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
& - \left(\tilde{\gamma}_2 - \tilde{\gamma}_3 \right) a_i^3 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + \left(\tilde{\gamma}_4 - \tilde{\gamma}_5 \right) a_i^2 \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 + 2\tilde{\gamma}_4 a_i^2 \nu^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& - \tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\alpha_B^{(i-1,i-1)} \right)^2 + \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 3\tilde{\alpha} a_i^2 \delta_{5,B}^{(i-1,i-1)} + \tilde{\alpha} a_i^2 \delta_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{4,B}^{(i-1,i-1)} \\
& - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{5,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{1,B}^{(i-1,i-1)} + \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)},
\end{aligned} \tag{42}$$

$$\begin{aligned}
\epsilon_{15,A}^{(i,i-1)} = & \epsilon_{15,B}^{(i-1,i-1)} + \tilde{\epsilon}_{15} a_i^5 \left(\sigma^{(i-1)} \right)^4 - (9\tilde{\epsilon}_{29} + 4\tilde{\epsilon}_{31} + \tilde{\epsilon}_{32} + 4\tilde{\epsilon}_{36} + \tilde{\epsilon}_{37} - 3\tilde{\epsilon}_{40} - 2\tilde{\epsilon}_{41} - \tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& - (12\tilde{\epsilon}_{43} + 9\tilde{\epsilon}_{44} + 5\tilde{\epsilon}_{45} + 12\tilde{\epsilon}_{46} + 9\tilde{\epsilon}_{47} + 5\tilde{\epsilon}_{48} - 3\tilde{\epsilon}_{49} - 5\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
& - (14\tilde{\epsilon}_{52} + 14\tilde{\epsilon}_{53} + 7\tilde{\epsilon}_{54} + 2\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 - 7\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 + \left(\tilde{\delta}_5 + \tilde{\delta}_7 - \tilde{\delta}_8 - 2\tilde{\delta}_9 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& - \left(2\tilde{\delta}_{11} + \tilde{\delta}_{12} - 2\tilde{\delta}_{14} \right) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - 2 \left(\tilde{\delta}_{10} + 2\tilde{\delta}_{11} - \tilde{\delta}_{13} - \tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& - \left(6\tilde{\delta}_{15} + 5\tilde{\delta}_{16} - \tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - \left(7\tilde{\delta}_{15} + \tilde{\delta}_{16} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 8\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} + (2\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
& - 8 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_4 + 2\tilde{\gamma}_5 \right) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& - (4\tilde{\gamma}_4 + 3\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 8 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} + 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
& - 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} - 5\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 5\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
& + 8\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)} + 2\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 - (4\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 \\
& - 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - 4\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{12,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} \\
& + \tilde{\beta} a_i \nu^{(i-1)} \delta_{12,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{43}$$

$$\begin{aligned}
\epsilon_{16,A}^{(i,i-1)} = & \epsilon_{16,B}^{(i-1,i-1)} + \tilde{\epsilon}_{16} a_i^5 \left(\sigma^{(i-1)} \right)^4 + (10\tilde{\epsilon}_{29} - 2\tilde{\epsilon}_{30} + 5\tilde{\epsilon}_{31} - \tilde{\epsilon}_{33} - 2\tilde{\epsilon}_{35} + 5\tilde{\epsilon}_{36} - \tilde{\epsilon}_{38} - 12\tilde{\epsilon}_{40} - 8\tilde{\epsilon}_{41} - 4\tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& + (13\tilde{\epsilon}_{43} + 10\tilde{\epsilon}_{44} + 5\tilde{\epsilon}_{45} + 13\tilde{\epsilon}_{46} + 10\tilde{\epsilon}_{47} + 5\tilde{\epsilon}_{48} - 14\tilde{\epsilon}_{49} - 11\tilde{\epsilon}_{50} - 24\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
& + (14\tilde{\epsilon}_{52} + 14\tilde{\epsilon}_{53} - 7\tilde{\epsilon}_{54} - 18\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 - 14\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 \\
& - \left(2\tilde{\delta}_5 + 3\tilde{\delta}_7 - 4\tilde{\delta}_8 - 8\tilde{\delta}_9 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + \left(5\tilde{\delta}_{11} - \tilde{\delta}_{13} - 8\tilde{\delta}_{14} \right) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + \left(5\tilde{\delta}_{10} + 9\tilde{\delta}_{11} - 7\tilde{\delta}_{12} - 10\tilde{\delta}_{13} - 8\tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + \left(15\tilde{\delta}_{15} + 8\tilde{\delta}_{16} - 11\tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + \left(16\tilde{\delta}_{15} - 7\tilde{\delta}_{16} - 7\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 24\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 12\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} - 5 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - \tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& + 12 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 8\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} + (3\tilde{\gamma}_4 + 7\tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& + 10 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 12 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
& - 32\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} + 16\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} + 17\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& + 15\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)} - 8\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 \\
& + (11\tilde{\gamma}_4 + 2\tilde{\gamma}_5) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 8\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& + 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 8\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - 2\tilde{\alpha} a_i^2 \delta_{10,B}^{(i-1,i-1)} + 5\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{13,B}^{(i-1,i-1)} \\
& + 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{10,B}^{(i-1,i-1)} - 5\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{13,B}^{(i-1,i-1)} - 16\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 11\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
& + 5\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{44}$$

$$\begin{aligned}
\epsilon_{17,A}^{(i,i-1)} = & \epsilon_{17,B}^{(i-1,i-1)} + \tilde{\epsilon}_{17} a_i^5 \left(\sigma^{(i-1)} \right)^4 - (6\tilde{\epsilon}_{29} + 3\tilde{\epsilon}_{31} + \tilde{\epsilon}_{34} + 3\tilde{\epsilon}_{36} + \tilde{\epsilon}_{39} - 9\tilde{\epsilon}_{40} - 6\tilde{\epsilon}_{41} - 3\tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& - (9\tilde{\epsilon}_{43} + 6\tilde{\epsilon}_{44} + 5\tilde{\epsilon}_{45} + 9\tilde{\epsilon}_{46} + 6\tilde{\epsilon}_{47} + 5\tilde{\epsilon}_{48} - 9\tilde{\epsilon}_{49} - 9\tilde{\epsilon}_{50} - 18\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
& - (14\tilde{\epsilon}_{52} + 14\tilde{\epsilon}_{53} - 6\tilde{\epsilon}_{54} - 15\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 + 7\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 \\
& + \left(\tilde{\delta}_5 + \tilde{\delta}_6 + 3\tilde{\delta}_7 - 3\tilde{\delta}_8 - 6\tilde{\delta}_9 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - \left(3\tilde{\delta}_{11} - 5\tilde{\delta}_{14} \right) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - \left(5\tilde{\delta}_{10} + 5\tilde{\delta}_{11} - 9\tilde{\delta}_{12} - 8\tilde{\delta}_{13} - 5\tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - \left(9\tilde{\delta}_{15} + 6\tilde{\delta}_{16} - 7\tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - \left(10\tilde{\delta}_{15} - 10\tilde{\delta}_{16} - 7\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 16\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} + 3(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 6(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
& - (\tilde{\gamma}_4 + 7\tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} - 3(\tilde{\gamma}_4 + 2\tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 6(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
& + 6(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} + 24\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} - 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
& - 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)} - (\tilde{\gamma}_2 - 6\tilde{\gamma}_3) a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 \\
& - 8\tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + (2\tilde{\gamma}_4 - 5\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 + 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& - 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - 3\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{14,B}^{(i-1,i-1)} \\
& + 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{14,B}^{(i-1,i-1)} + 13\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 9\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{45}$$

$$\begin{aligned}
\epsilon_{18,A}^{(i,i-1)} = & \epsilon_{18,B}^{(i-1,i-1)} + \tilde{\epsilon}_{18} a_i^5 \left(\sigma^{(i-1)} \right)^4 + (6\tilde{\epsilon}_{29} - \tilde{\epsilon}_{30} - 3\tilde{\epsilon}_{32} - \tilde{\epsilon}_{33} + 3\tilde{\epsilon}_{36} + 8\tilde{\epsilon}_{40} + 6\tilde{\epsilon}_{41} + 3\tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& + (4\tilde{\epsilon}_{43} + \tilde{\epsilon}_{44} + 9\tilde{\epsilon}_{46} + 6\tilde{\epsilon}_{47} + 3\tilde{\epsilon}_{48} + 8\tilde{\epsilon}_{49} + 12\tilde{\epsilon}_{50} + 15\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
& + (4\tilde{\epsilon}_{52} + 9\tilde{\epsilon}_{53} + 19\tilde{\epsilon}_{54} + 22\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 + 25\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 + \left(\tilde{\delta}_6 - 3\tilde{\delta}_8 - 6\tilde{\delta}_9 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& - \left(2\tilde{\delta}_{10} + \tilde{\delta}_{11} - 6\tilde{\delta}_{12} - 2\tilde{\delta}_{13} - 6\tilde{\delta}_{14} \right) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - \left(\tilde{\delta}_{10} - \tilde{\delta}_{11} - 15\tilde{\delta}_{12} - 9\tilde{\delta}_{13} - 5\tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& - \left(5\tilde{\delta}_{15} - 10\tilde{\delta}_{16} - 17\tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} + \left(\tilde{\delta}_{15} + 18\tilde{\delta}_{16} + 13\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& + 35\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} + 16\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 6\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} \\
& - 3(\tilde{\gamma}_2 - 2\tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 3\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + 6(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 6\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
& - 3\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} + 3(2\tilde{\gamma}_4 - \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 3\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& - 6(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} + 24\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
& - 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)} + 6\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 \\
& + (3\tilde{\gamma}_4 + 2\tilde{\gamma}_5) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 + 11\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& + 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 + 3\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{5,B}^{(i-1,i-1)} \\
& - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} + 12\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - 12\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
& - 5\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{46}$$

$$\begin{aligned}
\epsilon_{19,A}^{(i,i-1)} = & \epsilon_{19,B}^{(i-1,i-1)} + \tilde{\epsilon}_{19} a_i^5 \left(\sigma^{(i-1)} \right)^4 - (12\tilde{\epsilon}_{29} - \tilde{\epsilon}_{30} + 4\tilde{\epsilon}_{31} - 3\tilde{\epsilon}_{32} + \tilde{\epsilon}_{34} - 2\tilde{\epsilon}_{35} + 5\tilde{\epsilon}_{36} + 6\tilde{\epsilon}_{40} + 5\tilde{\epsilon}_{41} + 2\tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& - (12\tilde{\epsilon}_{43} + 9\tilde{\epsilon}_{44} + 4\tilde{\epsilon}_{45} + 13\tilde{\epsilon}_{46} + 12\tilde{\epsilon}_{47} + 5\tilde{\epsilon}_{48} + 5\tilde{\epsilon}_{49} + 12\tilde{\epsilon}_{50} + 9\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
& - (12\tilde{\epsilon}_{52} + 13\tilde{\epsilon}_{53} + 22\tilde{\epsilon}_{54} + 23\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 - 26\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 \\
& - (2\tilde{\delta}_5 + 5\tilde{\delta}_6 - 2\tilde{\delta}_8 - 4\tilde{\delta}_9) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + (4\tilde{\delta}_{10} + \tilde{\delta}_{11} - 6\tilde{\delta}_{12} - 2\tilde{\delta}_{14}) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + (7\tilde{\delta}_{10} + \tilde{\delta}_{11} - 17\tilde{\delta}_{12} - 4\tilde{\delta}_{13} + \tilde{\delta}_{14}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + (7\tilde{\delta}_{15} - 8\tilde{\delta}_{16} - 9\tilde{\delta}_{17}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + (5\tilde{\delta}_{15} - 15\tilde{\delta}_{16} - 7\tilde{\delta}_{17}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 15\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 12\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} + (5\tilde{\gamma}_2 - 4\tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& - 12(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 3(\tilde{\gamma}_4 + 2\tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& - (10\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 4(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + 12(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
& - 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} + 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + 12\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)} + 2(\tilde{\gamma}_2 - 2\tilde{\gamma}_3) a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 \\
& - (10\tilde{\gamma}_4 + 6\tilde{\gamma}_5) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2(2\tilde{\gamma}_4 - \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 - 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& - 22\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 2\tilde{\alpha} a_i^2 \delta_{10,B}^{(i-1,i-1)} - 5\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{10,B}^{(i-1,i-1)} \\
& + 5\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} - 10\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
& + 10\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{47}$$

$$\begin{aligned}
\epsilon_{20,A}^{(i,i-1)} = & \epsilon_{20,B}^{(i-1,i-1)} + \tilde{\epsilon}_{20} a_i^5 \left(\sigma^{(i-1)} \right)^4 + (8\tilde{\epsilon}_{29} - \tilde{\epsilon}_{30} + 3\tilde{\epsilon}_{31} - \tilde{\epsilon}_{32} + 4\tilde{\epsilon}_{36} - 6\tilde{\epsilon}_{40} - 4\tilde{\epsilon}_{41} - 3\tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& + (10\tilde{\epsilon}_{43} + 7\tilde{\epsilon}_{44} + 3\tilde{\epsilon}_{45} + 12\tilde{\epsilon}_{46} + 8\tilde{\epsilon}_{47} + 4\tilde{\epsilon}_{48} - 7\tilde{\epsilon}_{49} - 5\tilde{\epsilon}_{50} - 13\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
& + (10\tilde{\epsilon}_{52} + 12\tilde{\epsilon}_{53} - 7\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 - 4\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 + (\tilde{\delta}_6 - \tilde{\delta}_7 + 2\tilde{\delta}_8 + 4\tilde{\delta}_9) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& + 2(\tilde{\delta}_{12} - 2\tilde{\delta}_{14}) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} + (2\tilde{\delta}_{11} - 4\tilde{\delta}_{12} - 6\tilde{\delta}_{13} - 5\tilde{\delta}_{14}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& + 2(\tilde{\delta}_{16} - 3\tilde{\delta}_{17}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} + 3(\tilde{\delta}_{15} - 2\tilde{\delta}_{16} - 2\tilde{\delta}_{17}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& - 16\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - 7\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 8\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} \\
& - (4\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 8(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} + (3\tilde{\gamma}_4 + 4\tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& + (8\tilde{\gamma}_4 + 5\tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 8(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} - 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
& + 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} + 11\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} + 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
& - 8\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)} - 4\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 + 2(3\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 4\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 \\
& + 10\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 + 4\tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{7,B}^{(i-1,i-1)} \\
& - 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} - 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{48}$$

$$\begin{aligned}
\epsilon_{21,A}^{(i,i-1)} = & \epsilon_{21,B}^{(i-1,i-1)} + \tilde{\epsilon}_{21} a_i^5 \left(\sigma^{(i-1)} \right)^4 - (2\tilde{\epsilon}_{29} + \tilde{\epsilon}_{31} + \tilde{\epsilon}_{33} + 2\tilde{\epsilon}_{34} + \tilde{\epsilon}_{36}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& - (3\tilde{\epsilon}_{43} + 3\tilde{\epsilon}_{44} + 3\tilde{\epsilon}_{45} + 3\tilde{\epsilon}_{46} + 2\tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + \tilde{\epsilon}_{50}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 - 3(2\tilde{\epsilon}_{52} + \tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 \\
& - 6\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 + (\tilde{\delta}_7 + \tilde{\delta}_8) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - (2\tilde{\delta}_{10} + \tilde{\delta}_{11}) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - (\tilde{\delta}_{10} + \tilde{\delta}_{11} + \tilde{\delta}_{12} + \tilde{\delta}_{13} + 2\tilde{\delta}_{14}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - (5\tilde{\delta}_{15} + \tilde{\delta}_{17}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - (2\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + 3\tilde{\delta}_{17}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 2\tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} + (\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
& - 3\tilde{\gamma}_4 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 2(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
& + 2\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)} \\
& - (3\tilde{\gamma}_4 + 2\tilde{\gamma}_5) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{11,B}^{(i-1,i-1)} \\
& + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{8,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{11,B}^{(i-1,i-1)} - \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{49}$$

$$\begin{aligned}
\epsilon_{22,A}^{(i,i-1)} = & \epsilon_{22,B}^{(i-1,i-1)} + \tilde{\epsilon}_{22} a_i^5 \left(\sigma^{(i-1)} \right)^4 + (\tilde{\epsilon}_{29} - \tilde{\epsilon}_{30} - 2\tilde{\epsilon}_{32} - \tilde{\epsilon}_{33} - \tilde{\epsilon}_{35} + 3\tilde{\epsilon}_{40} + 2\tilde{\epsilon}_{41} + \tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& - (\tilde{\epsilon}_{43} + \tilde{\epsilon}_{44} + \tilde{\epsilon}_{45} + \tilde{\epsilon}_{46} - \tilde{\epsilon}_{47} - 2\tilde{\epsilon}_{49} - 4\tilde{\epsilon}_{50} - 4\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 - (2\tilde{\epsilon}_{52} + \tilde{\epsilon}_{53} - 4\tilde{\epsilon}_{54} - 6\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 \\
& + 5\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 + \left(\tilde{\delta}_5 + \tilde{\delta}_6 - 2\tilde{\delta}_7 - \tilde{\delta}_8 - \tilde{\delta}_9 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - \left(\tilde{\delta}_{10} - 2\tilde{\delta}_{14} \right) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + \left(4\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - \left(\tilde{\delta}_{15} - \tilde{\delta}_{16} - 4\tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + 3 \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 9\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} + 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& + \tilde{\gamma}_1 a_i^4 \gamma_{6,B}^{(i-1,i-1)} + (\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + (\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \nu^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - (3\tilde{\gamma}_4 + 2\tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
& - \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 2 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} + 7\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
& - 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} \\
& - \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)} - (\tilde{\gamma}_2 - \tilde{\gamma}_3) a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 + (5\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 2\tilde{\gamma}_4 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 \\
& + 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 + 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - \tilde{\alpha} a_i^2 \delta_{10,B}^{(i-1,i-1)} \\
& + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{9,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{10,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - \tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{50}$$

$$\begin{aligned}
\epsilon_{23,A}^{(i,i-1)} = & \epsilon_{23,B}^{(i-1,i-1)} + \tilde{\epsilon}_{23} a_i^5 \left(\sigma^{(i-1)} \right)^4 - (\tilde{\epsilon}_{35} + 3\tilde{\epsilon}_{36} + 3\tilde{\epsilon}_{37} + \tilde{\epsilon}_{38} + 7\tilde{\epsilon}_{40} + 4\tilde{\epsilon}_{41} + 2\tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& - (5\tilde{\epsilon}_{46} + 5\tilde{\epsilon}_{47} + 3\tilde{\epsilon}_{48} + 7\tilde{\epsilon}_{49} + 11\tilde{\epsilon}_{50} + 14\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 - (5\tilde{\epsilon}_{53} + 15\tilde{\epsilon}_{54} + 18\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 \\
& - 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 + \left(2\tilde{\delta}_8 + 5\tilde{\delta}_9 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 2 \left(2\tilde{\delta}_{12} + \tilde{\delta}_{13} + 2\tilde{\delta}_{14} \right) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - \left(10\tilde{\delta}_{12} + 7\tilde{\delta}_{13} + 4\tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - \left(8\tilde{\delta}_{16} + 12\tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - \left(12\tilde{\delta}_{16} + 9\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 26\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - 11\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 6\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} + 4\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
& + 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} + 6\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
& + 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
& + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{5,B}^{(i-1,i-1)} - 4\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 + 4\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 - 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& + 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{5,B}^{(i-1,i-1)} - 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} \\
& + 6\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{51}$$

$$\begin{aligned}
\epsilon_{24,A}^{(i,i-1)} = & \epsilon_{24,B}^{(i-1,i-1)} + \tilde{\epsilon}_{24} a_i^5 \left(\sigma^{(i-1)} \right)^4 - (\tilde{\epsilon}_{35} - \tilde{\epsilon}_{36} - 3\tilde{\epsilon}_{37} + \tilde{\epsilon}_{39} - 9\tilde{\epsilon}_{40} - 5\tilde{\epsilon}_{41} - 3\tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& + (\tilde{\epsilon}_{46} + 3\tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + 8\tilde{\epsilon}_{49} + 13\tilde{\epsilon}_{50} + 18\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 + (\tilde{\epsilon}_{53} + 16\tilde{\epsilon}_{54} + 21\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 \\
& + 19\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 - 3 \left(\tilde{\delta}_8 + 3\tilde{\delta}_9 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + \left(6\tilde{\delta}_{12} + 2\tilde{\delta}_{13} + 4\tilde{\delta}_{14} \right) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + \left(18\tilde{\delta}_{12} + 12\tilde{\delta}_{13} + 6\tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + \left(10\tilde{\delta}_{16} + 14\tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + \left(21\tilde{\delta}_{16} + 15\tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 30\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + 18\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 6\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 5\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} - 3\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} \\
& - 6\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} - 10\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
& - 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + 6\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 + \tilde{\gamma}_5 a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& - 6\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 + 10\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{6,B}^{(i-1,i-1)} + 12\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} \\
& + 6\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)} - 6\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{52}$$

$$\begin{aligned}
\epsilon_{25,A}^{(i,i-1)} = & \epsilon_{25,B}^{(i-1,i-1)} + \tilde{\epsilon}_{25} a_i^5 \left(\sigma^{(i-1)} \right)^4 - (\tilde{\epsilon}_{35} + \tilde{\epsilon}_{36} + \tilde{\epsilon}_{37} + \tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& - (2\tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + \tilde{\epsilon}_{49} + \tilde{\epsilon}_{50} + 2\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 - 2(\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 \\
& - 3\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 + (\tilde{\delta}_8 + \tilde{\delta}_9) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 4\tilde{\delta}_{12} a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - 2 \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 4 \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - 3 \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 8\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& - 2\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} + 2\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
& + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\tilde{\gamma}_5 a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& - 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 - 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 2\tilde{\beta}_i a_i \sigma^{(i-1)} \delta_{7,B}^{(i-1,i-1)} + 2\tilde{\beta}_i a_i \alpha_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)},
\end{aligned} \tag{53}$$

$$\begin{aligned}
\epsilon_{26,A}^{(i,i-1)} = & \epsilon_{26,B}^{(i-1,i-1)} + \tilde{\epsilon}_{26} a_i^5 \left(\sigma^{(i-1)} \right)^4 - (\tilde{\epsilon}_{38} + 2\tilde{\epsilon}_{39} - 6\tilde{\epsilon}_{40} - 4\tilde{\epsilon}_{41} - 2\tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& - (\tilde{\epsilon}_{47} + 2\tilde{\epsilon}_{48} - 6\tilde{\epsilon}_{49} - 8\tilde{\epsilon}_{50} - 14\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 - (3\tilde{\epsilon}_{53} - 10\tilde{\epsilon}_{54} - 15\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 \\
& + 13\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 - 4 \left(\tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 2 \left(\tilde{\delta}_{13} - 2\tilde{\delta}_{14} \right) a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + 8 \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 4 \left(\tilde{\delta}_{16} - 2\tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + 12 \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} + 22\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& + 16\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 4\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} - 6\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} + 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
& - 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} - 12\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} + 4\tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 + 4\tilde{\gamma}_5 a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& - 4\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 + 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 + 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& - 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{8,B}^{(i-1,i-1)} + 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} + 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{54}$$

$$\begin{aligned}
\epsilon_{27,A}^{(i,i-1)} = & \epsilon_{27,B}^{(i-1,i-1)} + \tilde{\epsilon}_{27} a_i^5 \left(\sigma^{(i-1)} \right)^4 - (2\tilde{\epsilon}_{37} + \tilde{\epsilon}_{38} + 9\tilde{\epsilon}_{40} + 6\tilde{\epsilon}_{41} + 3\tilde{\epsilon}_{42}) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \\
& - (2\tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + 9\tilde{\epsilon}_{49} + 12\tilde{\epsilon}_{50} + 17\tilde{\epsilon}_{51}) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 - (\tilde{\epsilon}_{53} + 15\tilde{\epsilon}_{54} + 20\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 \\
& - 19\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 + (\tilde{\delta}_8 + 4\tilde{\delta}_9) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_{B,i-1}^3 - 4\tilde{\delta}_{14} a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - (8\tilde{\delta}_{12} + 5\tilde{\delta}_{13} + 2\tilde{\delta}_{14}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_{B,i-1}^2 - 8\tilde{\delta}_{17} a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - (9\tilde{\delta}_{16} + 6\tilde{\delta}_{17}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_{B,i-1}^2 - 20\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - 7\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_{B,i-1}^2 \quad (55) \\
& + 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} \\
& + 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 4\tilde{\gamma}_3 a_i^3 \left(\beta_{B,i-1}^{(i-1,i-1)} \right)^2 + 4\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_{B,i-1}^{(i-1,i-1)} \right)^2 - 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 \\
& + 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_{B,i-1}^{(i-1,i-1)} \right)^2 - 2\tilde{\beta}_9 a_i \sigma^{(i-1)} \delta_{9,B}^{(i-1,i-1)} - 8\tilde{\beta}_9 a_i \beta_{B,i-1}^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 6\tilde{\beta}_9 a_i \beta_{B,i-1}^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned}$$

$$\begin{aligned}
\epsilon_{28,A}^{(i,i-1)} = & \epsilon_{28,B}^{(i-1,i-1)} + \tilde{\epsilon}_{28} a_i^5 \left(\sigma^{(i-1)} \right)^4 - 2 \left(\tilde{\epsilon}_{40} + \tilde{\epsilon}_{41} + \tilde{\epsilon}_{42} \right) a_i^4 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 - \left(3\tilde{\epsilon}_{49} + 3\tilde{\epsilon}_{50} + 5\tilde{\epsilon}_{51} \right) a_i^3 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^4 \\
& - \left(4\tilde{\epsilon}_{54} + 6\tilde{\epsilon}_{55} \right) a_i^2 \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^4 - 7\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^4 \left(\sigma^{(i-1)} \right)^4 + \left(\tilde{\delta}_8 + \tilde{\delta}_9 \right) a_i^4 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& - 2\tilde{\delta}_{14} a_i^3 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - 2 \left(\tilde{\delta}_{12} + \tilde{\delta}_{13} + \tilde{\delta}_{14} \right) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 4\tilde{\delta}_{17} a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} \\
& - 3 \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} - 10\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \alpha_B^{(i-1,i-1)} - 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^2 \beta_B^{(i-1,i-1)} \\
& - \tilde{\gamma}_3 a_i^3 \left(\beta_B^{(i-1,i-1)} \right)^2 + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{1,B}^{(i-1,i-1)} \\
& + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{2,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{3,B}^{(i-1,i-1)} - \tilde{\gamma}_5 a_i^2 \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& + \tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 - 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\alpha_B^{(i-1,i-1)} \right)^2 - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& + \tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - 2\tilde{\beta}_i \alpha_i \beta_B^{(i-1,i-1)} \gamma_{2,B}^{(i-1,i-1)} - 2\tilde{\beta}_i \alpha_i \beta_B^{(i-1,i-1)} \gamma_{3,B}^{(i-1,i-1)},
\end{aligned} \tag{56}$$

$$\begin{aligned}
\epsilon_{33,A}^{(i,i-1)} = & \epsilon_{33,A}^{(i-1,i-1)} + \tilde{\epsilon}_{33,A}^4 \left(\sigma^{(i-1)} \right)^5 - (3\tilde{\epsilon}_{43} - 2\tilde{\epsilon}_{44} + 3\tilde{\epsilon}_{45} + 3\tilde{\epsilon}_{46} + 3\tilde{\epsilon}_{48}) a_3^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^5 \\
& - (19\tilde{\epsilon}_{52} + 18\tilde{\epsilon}_{53} + 3\tilde{\epsilon}_{54} - 2\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^5 - 19\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^5 - \left(9\tilde{\delta}_{10} + 3\tilde{\delta}_{11} - \tilde{\delta}_{13} \right) a_i^3 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} \\
& - \left(11\tilde{\delta}_{15} + 3\tilde{\delta}_{17} \right) a_i^2 \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} - \left(18\tilde{\delta}_{15} - 5\tilde{\delta}_{16} + 3\tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} \\
& - 26\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} - 19\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} + 3 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
& + 3 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} + 3 \left(\tilde{\gamma}_4 + 2\tilde{\gamma}_5 \right) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} - 6 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
& + 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{3,B}^{(i-1,i-1)} + 9\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} + 15\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
& - 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + \left(9\tilde{\gamma}_4 - 2\tilde{\gamma}_5 \right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 + \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& + 5\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 - 3\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} - 3\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{13,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} \\
& + 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} - 7\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{61}$$

$$\begin{aligned}
\epsilon_{34,A}^{(i-1,i-1)} = & \epsilon_{34,B}^{(i-1,i-1)} + \tilde{\epsilon}_{34} a_i^4 \left(\sigma^{(i-1)} \right)^5 + (3\tilde{\epsilon}_{43} + 4\tilde{\epsilon}_{45} + 3\tilde{\epsilon}_{46} + 2\tilde{\epsilon}_{48}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^5 + (18\tilde{\epsilon}_{52} + 13\tilde{\epsilon}_{53} + 3\tilde{\epsilon}_{54}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^5 \\
& + 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^5 + (5\tilde{\delta}_{10} + 2\tilde{\delta}_{11} + \tilde{\delta}_{14}) a_i^3 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} + (9\tilde{\delta}_{15} + 2\tilde{\delta}_{17}) a_i^2 \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} \\
& + (11\tilde{\delta}_{15} - \tilde{\delta}_{16} + 4\tilde{\delta}_{17}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} + 19\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} + 16\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} \\
& - 3(\tilde{\gamma}_2 + \tilde{\gamma}_3) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 2(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} - 4(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
& + 6(\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{2,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{3,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
& - 12\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} + 9\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - 7\tilde{\gamma}_4 a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 \\
& - 7\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 + 3\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} + 2\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{14,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} \\
& - 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{62}$$

$$\begin{aligned}
\epsilon_{35,A}^{(i,i-1)} = & \epsilon_{35,B}^{(i-1,i-1)} + \tilde{\epsilon}_{35} a_i^4 \left(\sigma^{(i-1)} \right)^5 - (2\tilde{\epsilon}_{43} + \tilde{\epsilon}_{45} + \tilde{\epsilon}_{46} + \tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} - \tilde{\epsilon}_{49} - 2\tilde{\epsilon}_{51}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^5 \\
& - (7\tilde{\epsilon}_{52} + 6\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} - \tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^5 - 5\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^5 \\
& - (2\tilde{\delta}_{10} + \tilde{\delta}_{11} + 2\tilde{\delta}_{12} - \tilde{\delta}_{13} - \tilde{\delta}_{14}) a_i^3 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} - (2\tilde{\delta}_{15} + 2\tilde{\delta}_{16} + \tilde{\delta}_{17}) a_i^2 \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} \\
& - (5\tilde{\delta}_{15} + \tilde{\delta}_{17}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} - 5\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} - 5\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} \\
& + 2\tilde{\gamma}_2 a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + (\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} + (2\tilde{\gamma}_4 + 3\tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
& - 2(2\tilde{\gamma}_4 + \tilde{\gamma}_5) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{2,B}^{(i-1,i-1)} + 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
& + 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + (3\tilde{\gamma}_4 - \tilde{\gamma}_5) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 \\
& - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + \tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 - 2\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} - \tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{10,B}^{(i-1,i-1)} \\
& + 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)},
\end{aligned} \tag{63}$$

$$\begin{aligned}
\epsilon_{36,A}^{(i,i-1)} = & \epsilon_{36,B}^{(i-1,i-1)} + \tilde{\epsilon}_{36} a_i^4 \left(\sigma^{(i-1)} \right)^5 + (3\tilde{\epsilon}_{46} + 2\tilde{\epsilon}_{47} + \tilde{\epsilon}_{48} + \tilde{\epsilon}_{50}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^5 + 3(\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + \tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^5 \\
& + \tilde{\delta}_{56} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^5 + \tilde{\delta}_{12} a_i^3 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} + 2 \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} \\
& + \left(\tilde{\delta}_{16} + \tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} + 8\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} \\
& + 4\tilde{\gamma}_3 a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} - 4\tilde{\gamma}_5 a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{3,B}^{(i-1,i-1)} \\
& + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} \\
& - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{11,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{64}$$

$$\begin{aligned}
\epsilon_{41,A}^{(i,i-1)} = & \epsilon_{41,B}^{(i-1,i-1)} + \tilde{\epsilon}_{41} a_i^4 \left(\sigma^{(i-1)} \right)^5 + (8\tilde{\epsilon}_{43} + 4\tilde{\epsilon}_{45} + 8\tilde{\epsilon}_{46} + 4\tilde{\epsilon}_{48} + \tilde{\epsilon}_{49} - \tilde{\epsilon}_{50} + \tilde{\epsilon}_{51}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^5 \\
& + (28\tilde{\epsilon}_{52} + 28\tilde{\epsilon}_{53} + 7\tilde{\epsilon}_{54} - 4\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^5 + 14\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^5 \\
& + (8\tilde{\delta}_{10} + 4\tilde{\delta}_{11} - 3\tilde{\delta}_{12} - 2\tilde{\delta}_{14}) a_i^3 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} + 2 \left(4\tilde{\delta}_{15} - \tilde{\delta}_{17} \right) a_i^2 \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} \\
& + \left(20\tilde{\delta}_{15} - 3\tilde{\delta}_{16} - 3\tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} - 2\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} + 4\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} \\
& - 8 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} - (4\tilde{\gamma}_4 + 7\tilde{\gamma}_5) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} - 8 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} \\
& + 16 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 2\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{2,B}^{(i-1,i-1)} - 20\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{3,B}^{(i-1,i-1)} \\
& - 18\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} - 24\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} + 24\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
& - 3 \left(4\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 - 30\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} - 18\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 + 8\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} \\
& + 4\tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} - 8\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} + 10\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
& + 8\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} + 8\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{69}$$

$$\begin{aligned}
\epsilon_{42,A}^{(i,i-1)} = & \epsilon_{42,B}^{(i-1,i-1)} + \tilde{\epsilon}_{42} a_i^4 \left(\sigma^{(i-1)} \right)^5 - (2\tilde{\epsilon}_{43} + \tilde{\epsilon}_{45} + 2\tilde{\epsilon}_{46} + \tilde{\epsilon}_{48} - \tilde{\epsilon}_{49} - \tilde{\epsilon}_{50} - \tilde{\epsilon}_{51}) a_i^3 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^5 \\
& - (7\tilde{\epsilon}_{52} + 7\tilde{\epsilon}_{53} - 5\tilde{\epsilon}_{55}) a_i^2 \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^5 + 7\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^3 \left(\sigma^{(i-1)} \right)^5 - \left(2\tilde{\delta}_{10} + \tilde{\delta}_{11} - \tilde{\delta}_{12} - \tilde{\delta}_{14} \right) a_i^3 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} \\
& + \left(3\tilde{\delta}_{17} - 2\tilde{\delta}_{15} \right) a_i^2 \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} - \left(5\tilde{\delta}_{15} - \tilde{\delta}_{16} - 2\tilde{\delta}_{17} \right) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} + 13\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \alpha_B^{(i-1,i-1)} \\
& + 2\tilde{\delta}_{18} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^3 \beta_B^{(i-1,i-1)} + 2 \left(\tilde{\gamma}_2 + \tilde{\gamma}_3 \right) a_i^3 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + \left(\tilde{\gamma}_4 + 2\tilde{\gamma}_5 \right) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} \\
& + 2 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} - 4 \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \nu^{(i-1)} \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} + 10\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{3,B}^{(i-1,i-1)} \\
& + 5\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{4,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{5,B}^{(i-1,i-1)} - 6\tilde{\gamma}_6 a_i \left(\nu^{(i-1)} \right)^2 \sigma^{(i-1)} \gamma_{6,B}^{(i-1,i-1)} \\
& + \left(3\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 + 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \alpha_B^{(i-1,i-1)} \beta_B^{(i-1,i-1)} + 5\tilde{\gamma}_6 a_i \nu^{(i-1)} \sigma^{(i-1)} \left(\beta_B^{(i-1,i-1)} \right)^2 - 2\tilde{\alpha} a_i^2 \delta_{15,B}^{(i-1,i-1)} \\
& - \tilde{\alpha} a_i^2 \delta_{17,B}^{(i-1,i-1)} + 2\tilde{\beta} a_i \nu^{(i-1)} \delta_{15,B}^{(i-1,i-1)} + \tilde{\beta} a_i \nu^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{4,B}^{(i-1,i-1)} \\
& - 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{5,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \alpha_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{70}$$

$$\begin{aligned}
\epsilon_{43,A}^{(i,i-1)} = & \epsilon_{43,B}^{(i-1,i-1)} + \tilde{\epsilon}_{43} a_i^3 \left(\sigma^{(i-1)} \right)^6 + (4\tilde{\epsilon}_{52} + 3\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} + 2\tilde{\epsilon}_{55}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^6 + 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^6 \\
& + 2\tilde{\delta}_{15} a_i^2 \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} + 11\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} - \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
& - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\beta_B^{(i-1,i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^2 + 3\tilde{\alpha} a_i^2 \delta_{18,B}^{(i-1,i-1)} \\
& + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{15,B}^{(i-1,i-1)} - 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{18,B}^{(i-1,i-1)} + \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{71}$$

$$\begin{aligned}
\epsilon_{44,A}^{(i,i-1)} = & \epsilon_{44,B}^{(i-1,i-1)} + \tilde{\epsilon}_{44} a_i^3 \left(\sigma^{(i-1)} \right)^6 - (4\tilde{\epsilon}_{52} + 3\tilde{\epsilon}_{53} + 7\tilde{\epsilon}_{55}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^6 - 60\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^6 \\
& - \left(2\tilde{\delta}_{15} - \tilde{\delta}_{16} \right) a_i^2 \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} - 43\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \alpha_B^{(i-1,i-1)} - 3\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} \\
& + 3\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} + 15\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} + 14\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 \\
& - 3\tilde{\alpha} a_i^2 \delta_{18,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{16,B}^{(i-1,i-1)} + 3\tilde{\beta} a_i \nu^{(i-1)} \delta_{18,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \gamma_{6,B}^{(i-1,i-1)} \beta_B^{(i-1,i-1)},
\end{aligned} \tag{72}$$

$$\begin{aligned}
\epsilon_{45,A}^{(i,i-1)} = & \epsilon_{45,B}^{(i-1,i-1)} + \tilde{\epsilon}_{45} a_i^3 \left(\sigma^{(i-1)} \right)^6 + (6\tilde{\epsilon}_{52} + \tilde{\epsilon}_{53} + 6\tilde{\epsilon}_{55}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^6 + 48\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^6 \\
& - \left(\tilde{\delta}_{15} - \tilde{\delta}_{17} \right) a_i^2 \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} + 31\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \alpha_B^{(i-1,i-1)} + 4\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} \\
& - \left(\tilde{\gamma}_4 + \tilde{\gamma}_5 \right) a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} + \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} - 10\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
& - 12\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 + \tilde{\alpha} a_i^2 \delta_{18,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{17,B}^{(i-1,i-1)} - \tilde{\beta} a_i \nu^{(i-1)} \delta_{18,B}^{(i-1,i-1)} + \tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{73}$$

$$\begin{aligned}
\epsilon_{46,A}^{(i,i-1)} = & \epsilon_{46,B}^{(i-1,i-1)} + \tilde{\epsilon}_{46} a_i^3 \left(\sigma^{(i-1)} \right)^6 + (\tilde{\epsilon}_{53} + \tilde{\epsilon}_{54} - 2\tilde{\epsilon}_{55}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^6 - 18\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^6 + \tilde{\delta}_{16} a_i^2 \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} \\
& - 14\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \alpha_B^{(i-1,i-1)} - \tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} + \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
& + 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{15,B}^{(i-1,i-1)},
\end{aligned} \tag{74}$$

$$\begin{aligned}
\epsilon_{47,A}^{(i,i-1)} &= \epsilon_{47,B}^{(i-1,i-1)} + \tilde{\epsilon}_{47} a_i^3 \left(\sigma^{(i-1)} \right)^6 - (\tilde{\epsilon}_{53} - 15\tilde{\epsilon}_{55}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^6 + 102\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^6 - 2\tilde{\delta}_{16} a_i^2 \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} \\
&+ 74\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \alpha_B^{(i-1,i-1)} + 2\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} - 4\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} - 24\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} \\
&- 8\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} - 24\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{16,B}^{(i-1,i-1)} + 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{75}$$

$$\begin{aligned}
\epsilon_{48,A}^{(i,i-1)} &= \epsilon_{48,B}^{(i-1,i-1)} + \tilde{\epsilon}_{48} a_i^3 \left(\sigma^{(i-1)} \right)^6 + (5\tilde{\epsilon}_{53} - 12\tilde{\epsilon}_{55}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^6 - 69\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^6 \\
&- (2\tilde{\delta}_{16} + \tilde{\delta}_{17}) a_i^2 \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} - 48\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \alpha_B^{(i-1,i-1)} - 3\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} \\
&- 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} + 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} + 19\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{17,B}^{(i-1,i-1)},
\end{aligned} \tag{76}$$

$$\begin{aligned}
\epsilon_{49,A}^{(i,i-1)} &= \epsilon_{49,B}^{(i-1,i-1)} + \tilde{\epsilon}_{49} a_i^3 \left(\sigma^{(i-1)} \right)^6 + (\tilde{\epsilon}_{54} + 7\tilde{\epsilon}_{55}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^6 + 42\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^6 \\
&- (\tilde{\delta}_{16} + 2\tilde{\delta}_{17}) a_i^2 \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} + 20\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \alpha_B^{(i-1,i-1)} - 9\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} \\
&- \tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} - 10\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} - 2\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} \\
&- 8\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2 + 2\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)},
\end{aligned} \tag{77}$$

$$\begin{aligned}
\epsilon_{50,A}^{(i,i-1)} &= \epsilon_{50,B}^{(i-1,i-1)} + \tilde{\epsilon}_{50} a_i^3 \left(\sigma^{(i-1)} \right)^6 + (3\tilde{\epsilon}_{54} - 6\tilde{\epsilon}_{55}) a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^6 - 63\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^6 + 2\tilde{\delta}_{16} a_i^2 \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} \\
&- 50\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \alpha_B^{(i-1,i-1)} - 2\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} + 2\tilde{\gamma}_5 a_i^2 \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} \\
&+ 16\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} + 4\tilde{\gamma}_6 a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^2 \gamma_{6,B}^{(i-1,i-1)} - 4\tilde{\beta} a_i \beta_B^{(i-1,i-1)} \gamma_{6,B}^{(i-1,i-1)} + 13\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2,
\end{aligned} \tag{78}$$

$$\begin{aligned}
\epsilon_{51,A}^{(i,i-1)} &= \epsilon_{51,B}^{(i-1,i-1)} + \tilde{\epsilon}_{51} a_i^3 \left(\sigma^{(i-1)} \right)^6 + 3\tilde{\epsilon}_{55} a_i^2 \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^6 + 21\tilde{\epsilon}_{56} a_i \left(\nu^{(i-1)} \right)^2 \left(\sigma^{(i-1)} \right)^6 + \tilde{\delta}_{17} a_i^2 \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} \\
&+ 20\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \alpha_B^{(i-1,i-1)} + 5\tilde{\delta}_{18} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^4 \beta_B^{(i-1,i-1)} - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{4,B}^{(i-1,i-1)} \\
&- 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{5,B}^{(i-1,i-1)} - 3\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^2 \left(\beta_B^{(i-1,i-1)} \right)^2,
\end{aligned} \tag{79}$$

$$\begin{aligned}
\epsilon_{52,A}^{(i,i-1)} &= \epsilon_{52,B}^{(i-1,i-1)} + \tilde{\epsilon}_{52} a_i^2 \left(\sigma^{(i-1)} \right)^7 + \tilde{\epsilon}_{56} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^7 + \tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \beta_B^{(i-1,i-1)} \\
&+ \tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)} + \tilde{\beta} a_i \sigma^{(i-1)} \delta_{18,B}^{(i-1,i-1)},
\end{aligned} \tag{80}$$

$$\begin{aligned}
\epsilon_{53,A}^{(i,i-1)} &= \epsilon_{53,B}^{(i-1,i-1)} + \tilde{\epsilon}_{53} a_i^2 \left(\sigma^{(i-1)} \right)^7 + 6\tilde{\epsilon}_{56} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^7 - 3\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \beta_B^{(i-1,i-1)} \\
&- 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)} - 2\tilde{\beta} a_i \sigma^{(i-1)} \delta_{18,B}^{(i-1,i-1)},
\end{aligned} \tag{81}$$

$$\epsilon_{54,A}^{(i,i-1)} = \epsilon_{54,B}^{(i-1,i-1)} + \tilde{\epsilon}_{54} a_i^2 \left(\sigma^{(i-1)} \right)^7 - 14\tilde{\epsilon}_{56} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^7 + 6\tilde{\delta}_{18} a_i \left(\sigma^{(i-1)} \right)^5 \beta_B^{(i-1,i-1)} + 6\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)}, \tag{82}$$

$$\epsilon_{55,A}^{(i,i-1)} = \epsilon_{55,B}^{(i-1,i-1)} + \tilde{\epsilon}_{55} a_i^2 \left(\sigma^{(i-1)} \right)^7 + 14\tilde{\epsilon}_{56} a_i \nu^{(i-1)} \left(\sigma^{(i-1)} \right)^7 - 5\tilde{\delta}_{18} a_i \beta_B^{(i-1,i-1)} \left(\sigma^{(i-1)} \right)^5 - 4\tilde{\gamma}_6 a_i \left(\sigma^{(i-1)} \right)^3 \gamma_{6,B}^{(i-1,i-1)}, \tag{83}$$

$$\epsilon_{56,A}^{(i,i-1)} = \epsilon_{56,B}^{(i-1,i-1)} + \tilde{\epsilon}_{56} a_i \left(\sigma^{(i-1)} \right)^8. \tag{84}$$